Electric Vehicle Management System Using Artificial Intelligence and Machine Learning.

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AI-Based EV Anomaly Detection

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

There is a growing demand for intelligent Electric Vehicle (EV) management solutions that go beyond conventional vehicle monitoring and reactive maintenance. Existing EV management systems primarily provide information such as battery status, charging progress, driving statistics, and diagnostic fault codes. However, these systems offer limited capabilities for detecting unknown operational anomalies, identifying early signs of component failures, and providing intelligent maintenance recommendations. As a result, many abnormalities remain unnoticed until they develop into critical failures, leading to unexpected vehicle breakdowns, increased maintenance costs, reduced vehicle reliability, and decreased user confidence. This highlights a significant research gap in applying Artificial Intelligence (AI) and Machine Learning (ML) to analyze operational vehicle data for proactive anomaly detection and predictive maintenance.

The proposed AI-Based EV Anomaly Detection System (AIEVADS) introduces an intelligent component within an Electric Vehicle Management System (EVMS) that utilizes AI and Machine Learning techniques to continuously analyze vehicle operational data, including battery voltage, battery current, battery temperature, motor temperature, state of charge, vehicle speed, acceleration, charging status, and power consumption. The research evaluates multiple anomaly detection models, namely Isolation Forest, One-Class Support Vector Machine (One-Class SVM), and Deep Auto Encoder Neural Networks, to identify abnormal operating patterns that may indicate potential mechanical or electrical failures before they become critical. The best-performing model will be integrated into a Flutter-based mobile application to provide vehicle health assessments, anomaly alerts, fault severity classification, maintenance recommendations, and historical analytics through an intuitive user interface.

This research aims to develop an intelligent and scalable predictive maintenance framework that improves vehicle reliability, minimizes unexpected failures, reduces maintenance costs, and supports data-driven maintenance decisions for EV owners and service providers. Furthermore, the proposed solution contributes to the advancement of AI-driven Electric

Vehicle Management Systems by enabling proactive vehicle health monitoring and promoting safer, more efficient, and sustainable electric transportation.

Keywords: Electric Vehicles (EV), Electric Vehicle Management System (EVMS), Artificial Intelligence (AI), Machine Learning (ML), Anomaly Detection, Predictive Maintenance, Isolation Forest, One-Class Support Vector Machine (One-Class SVM), Deep Auto encoder, Vehicle Health Monitoring.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
AI/ML	Artificial Intelligence / Machine Learning
AIEVADS	AI-Based EV Anomaly Detection System
API	Application Programming Interface
DSR	Design Science Research
EV	Electric Vehicle
EVMS	Electric Vehicle Management System
F1-Score	Harmonic Mean of Precision and Recall
GPU	Graphics Processing Unit
HTTPS	Hypertext Transfer Protocol Secure
LKR	Sri Lankan Rupee
ML	Machine Learning
PII	Personally Identifiable Information
REST	Representational State Transfer
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
SoC (SOC)	State of Charge
SSD	Solid-State Drive
SVM	Support Vector Machine
UI	User Interface

1. INTRODUCTION

1.1 Overall project context

The progress made in Electric Vehicles (EVs) has emerged as a significant movement paving the way for sustainable transportation, which reduces reliance on fossil fuels and curbs environmental pollution. Given that EVs are increasingly being adopted, managing and monitoring of these vehicles properly is necessary to make sure that they are operated safely and efficiently and that users' expectations are met. However, conventional ways of monitoring EVs are not very effective in detecting unusual operational situations and giving timely diagnostic assistance. Recent developments in Artificial Intelligence (AI) and Machine Learning (ML) have brought possibilities for improving EV monitoring by means of data analysis and anomaly detection. The emergence of these technologies enables the analysis of vehicle operational indicators.

The aim of SmartEV – Electric Vehicle Management System that makes use of artificial intelligence and machine Learning is to increase the monitoring and management of electric vehicles by utilizing smart data analysis. While working as a separate independent research component, AI-based EV Anomaly Detection System involves the analysis of parameters for detecting various anomalous operational conditions. Significant parameters for this purpose include temperature of the battery, temperature of the vehicle, battery voltage and current, data related to the motor in terms of its heat, speed of the vehicle, acceleration, consumption of power and the level of charge. Various machine learning technologies are implemented for determining deviation from regular operations and assist in identification of vehicle defects. All the detected anomalies can be presented in the form of detailed information and diagnosis via a web-based interface, which allows users to evaluate the functioning of the vehicle.

1.2 Individual component context

The AI-based EV anomaly detection system is a separate part of the SmartEV system, which is the electric vehicle management system based on AI and ML technologies. The objective of this component is to determine when there is something wrong with the functioning of electric

vehicles and also gather information regarding the origin of the problem through an advanced and internet-based system. The system monitors the operational data of electric vehicles to identify different types and patterns that can suggest possible problems with the vehicles. The parametric input features comprise the battery voltage, current, battery temperature, motor temperature, speed of the car, acceleration, consumption of energy, SoC and the state of the charging process.

To help analyze the vehicle's behavior, machine learning mechanisms are utilized to detect any deviation from the vehicle's usual activities. The anomaly detection mechanism focuses on identifying anomalies in relation to battery operation, overtemperature, electrical changes in the vehicle, and the motor's operation. The anomalies discovered are further employed to assist in any diagnosis and investigations.

The machine learning model used for anomaly detection is combined into the SmartEV system. During the process of detection, the model processes the existing data from the vehicles. The results of the detection process are displayed in a web-based dashboard, where people can keep track of the vehicle's condition and detect any possible anomalies.

Accordingly, this particular component includes vehicle operation data analysis, anomaly detection using machine learning, and visualization via an online dashboard in order to provide sophisticated assistance in monitoring and verifying the condition of electric vehicles. This component can be distinguished from the other components in terms of its short-term focus on anomaly detection, whereas other components are designed for predicting the long-term state of health (SoH) of batteries.

1.3 Problem statement

Electric vehicles (EVs) are becoming more vital in the global transportation sector due to their ability to mitigate pollution and reliance on fossil fuel consumption. Nonetheless, guaranteeing the long-term operational safety of EVs remains a major challenge since EV performance relies on various dependent electrical, thermal, battery, and motor-related variables since deviations in these variables could signal possible technical challenges that are capable of impacting vehicle operation and reliability. While traditional EV monitoring methods may employ predetermined thresholds or simple monitoring systems to determine the existence of anomalous operational

conditions, such techniques might not provide full coverage against risk of identifying sophisticated abnormal behaviors that involve a number of vehicle's variables. In addition, without intelligent analysis of operational data, it becomes difficult for users to detect early signs of possible faults.

The use of Machine Learning (ML) techniques enables the improvement of EV monitoring by using operation data and finding anomalies. Nevertheless, a good anomaly detection approach needs to consider various vehicle parameters, such as the voltage of the battery, current, battery temperature, motor temperature, vehicle speed, acceleration, power usage, state of charge (SoC), and charging conditions. The analysis of these parameters together allows for discovering abnormal operational factors that can hardly be detected through the analysis of the individual measurements.

Thus, the issue that is addressed within the scope of this research section is related to the unavailability of an intelligent method of detecting abnormal operating conditions of EVs based on the analysis of several vehicle parameters and provision of useful diagnostics to the users. Without effective detection of anomalies, possible issues concerning the vehicle's electrical system, temperature, motor, or charging system will remain unresolved, leaving people unable to address the problems effectively.

To solve this issue, the AI-Based EV Anomaly Detection System which has been proposed and designed as a separate section of SmartEV - Electric Vehicle Management System Using AI and ML, employs machine learning for analyzing the operation data of the vehicle in order to identify deviations from the normal operation behavior. The system monitors variables such as voltage, current, temperature, speed, acceleration, power consumption, state of charge, and charging mode. The identified anomalies are then available through the web-based platform and the dashboard of the system.

In this way, this part of the study is graced by the development of a data-based technology for the anomaly detection in operations of electric vehicles. The result of this technology is the enhancement of vehicle monitoring, promotion of timely detection of potential problems, and provision of assistance in making conscious choices related to the maintenance and safety of the vehicles.

1.4 Research gap

Earlier studies have looked into the usage of Machine Learning techniques for monitoring of electric vehicle batteries, diagnosing faults, making forecasts, analyzing battery operations and detecting abnormalities. It has been shown by these studies that intelligent technologies can help in detecting abnormal behavior of vehicle, making fault detection process easier and gathering data for the purposes of maintenance and decision making regarding the operations of EVs.

Despite it, existing EV monitoring system may not successfully detect complex operation anomalies which involve a number of vehicle parameters. Systems that rely on certain thresholds and individual readings of a single sensor may miss anomalies which happen at once across electrical, thermal, motor and charging parameters. Moreover, some studies tend to concentrate on long-term battery SoH estimation and battery depreciation assessment or remaining mileage instead of focusing on short-term abnormal operations indicating urgent technical problem.

Moreover, efficient EV anomaly detection necessitates the examination of many interconnected parameters, such as battery voltage, current, battery temperature, motor temperature, vehicle speed, acceleration, power consumption, State of Charge (SoC), and charging state. It is essential to have a unified method that analyzes these operational parameters, detects deviations from the standard pattern and visualizes the identified anomalies in an easy-to-use interface. In the absence of such method, users might struggle to understand the data on vehicle performance and identify abnormal cases that need investigation.

Thus, the research gap covered by this particular research component is the requirement for a unified machine learning approach to analyzing several EV operational parameters aimed at short-term anomalies detection and reporting the results through the web-based monitoring system. The AI-based EV Anomalies Detection System that is proposed fulfills this requirement of combining the analysis of operational data from the vehicle, anomaly detection based on machine learning, backend integration, and visualization by means of the dashboard in the framework of the SmartEV – Electric Vehicle Management System Using AI and ML.

This component is designed to help identify possibly abnormal vehicle states and provide useful information for subsequent diagnostic analysis and maintenance decisions by focusing on short-term operational irregularities instead of predicting the health of batteries in the long run.

1.5 Aim and objectives

1.5.1 Aim

The purpose of this research is to create the AI-Based EV Anomaly Detection System, which will be an important part of SmartEV – Electric Vehicle Management System Utilizing AI and ML. The system will be based on machine learning and other advanced methods for processing the data obtained from operation of electric vehicles to identify anomalies in their operation. The goal is to help improve monitoring of electric vehicles through the detection of possible malfunctions related to electricity, heat, motors, batteries, and charge, as well as visualization of the results using the web dashboard for investigating, safekeeping, and making decisions on maintenance of electric vehicles.

1.5.2 Objectives

1. Select the essential electric vehicle operating variables such as voltage and current of the battery, battery and motor temperatures, vehicle speed and acceleration, power use, and state of charge for anomaly detection.
2. Which refers to preparing and cleaning the approved operational data of electric vehicles by ensuring it is the right data for the anomaly detection process of machine learning.
3. Create and use a model of machine learning-based detection of anomalies.
4. Link the created model of anomaly detection of the SmartEV web application by modifying its back end, which will process the data from the vehicle and recall the anomalies.
5. Create a website hub that will provide the anomalies detected and data related to the operation of vehicles in a simple and understandable way for users.
6. Test the performance of the anomaly detection model by evaluating it with quality measurements.

1.6 Research Questions or Hypotheses

Research Questions:

1. What are the key parameters of electric vehicles that can contribute to identifying unusual working conditions of vehicles?
2. To what extent can the data-related machine learning techniques be employed to detect the abnormalities in the operation of electric vehicles taking into consideration electrical, thermal, motor, and characteristics connected to charging?
3. How well can the developed anomaly detection model differentiate between normal functional behavior and operation of electric vehicles?

4. What are the technical opportunities for integrating the machine learning detection system into the web-based management of electric vehicles?
5. To what extent can the proposed service contribute to electric vehicle monitoring, technical problems identification, and maintenance decision-making?

Hypotheses:

- H_0 (Null Hypothesis): The generated machine learning-based system of anomaly detecting gives no effective solution to identify abnormal conditions of an electric vehicle based on operational data analysis.
- H_1 (Alternative Hypothesis): The generated machine learning-based system of anomaly detecting gives an effective solution to identify abnormal conditions of an electric vehicle based on operational data analysis.

1.7 Scope, Limitations and Assumptions

Scope

The AI-Based EV Anomaly Detection System is one of the key elements in the comprehensive SmartEV – Electric Vehicle Management System, which is built on the foundations of AI and ML technology. This module deals with analysis of EV functioning predictors, determining abnormality of EV functioning patterns, and delivering the results of anomaly detection in terms of vehicle monitoring and diagnostics.

The objective of this research component consists of:

- Finding applicable EV operational properties for anomaly detection.
- Gathering or taking EV operational information from a suitable set of data.
- Pre-treating and adjusting the collected data accordingly to machine learning.
- Carefully considering performance indicators such as voltage, current, temperature of batteries, temperature of motors, speed, acceleration, power consumption, state of charge, and process of charging.
- Creating and using a machine-learning anomaly detection model.
- Observing any odd behavior concerning traditional EV operation.
- Connecting an anomaly recognition model with Smart EV's backend.
- Showing results of an anomaly recognition using a web dashboard.
- Measuring the effectiveness of the model and the progress of the interactions between the model and web app.

- Providing data that can help users with identifying possibly wrong conditions of vehicles and making decisions about further technical investigation.

This element emphasizes short-term anomaly detection and support for diagnostics in electric vehicles instead of long-term battery state of affairs forecasting.

Limitations

- Availability and quality of the dataset: The performance of the anomaly detection model is largely reliant on the quality, completeness, and representativity of the selected dataset. The presence of missing values, erroneous data, imbalanced classes, and scarcity of abnormal cases may reach negative impact on the efficiency of the model to find anomalies.
- Insufficient validation on real conditions: In case the model is trained and tested on publicly available or simulated data, its performance can hardly correspond to the actual conditions of EV operation. There can be a necessity of real-life testing with different types of EVs and in various operating conditions before applying the model in practice.
- Restrictions in operational parameters: The model can only examine factors represented in the chosen dataset. Variables such as the battery condition, component aging, environmental temperature, road quality, driving habits, and car characteristics could have a role in the workings of the vehicle but may not necessarily be included in the existing data.
- Limitations of machine learning: It is possible for machine learning systems to exhibit false positives, which refers to normal phenomena unreasonably recognized as an anomaly, and false negatives, which imply failure in the detection of genuine anomalies. Therefore, the outcome of the model needs to be assessed with the help of relevant metrics like precision, recall, F1-score, etc.
- Diagnostic Constraints: While anomaly detection can highlight abnormal activity, it doesn't mean it will find out the reasons for malfunctioning of a car. It has to be backed up by additional tests and professional technical analysis to identify the root cause.
- System Integration Drawbacks: Accessibility of operational data, backend connection, processing capabilities, and communication on a dashboard may impact system activity.
- Safety restrictions: The proposed solution is meant to allow for monitoring and diagnostic examination. The predictions should not be considered an alternative to the manufacturer’s diagnostic systems, skilled technicians or recognized EV safety processes.

Assumptions

The study is formulated on the basis of several presumptions:

1. The chosen dataset or information source contains corresponding EV operational parameters, which can effectively assist with recognizing abnormal operating conditions.

2. The accessible operational data has enough representativeness, which enables to create and evaluate the model.
3. The input parameters that were selected provide some insights into the EV behaviors related to electrical, thermal, motor, battery, and charging conditions that were abnormal.
4. The methods of data processing and feature preparation can produce inputs in compliance with the selected algorithm.
5. The anomaly detection model developed can be integrated with the backend of SmartEV to process vehicle data and provide the results of detection.
6. The web-based dashboard provides the results of the model in a comprehensible form for the users to facilitate their investigation.
7. Users will use the anomaly detection results for their decision-making processes without considering them as definitive proof of the vehicle malfunction.
8. The performance results obtained during the evaluation will serve as an indicator of the model efficacy under the conditions depicted in the evaluation dataset.

1.8 Individual Contribution

This thesis outlines the unique contribution made by Vidushan A A D D N (IT22157928) to the research project R26-IT-029, given the title Electric Vehicle Management System Using AI and ML, devised under the SmartEV project.

The research study in question encompasses the use of Artificial Intelligence (AI) and Machine Learning (ML) to transform the management of electric vehicles. In this research endeavor, the individual contribution is aimed at the creation of the AI-Based EV Anomaly Detection System that is designed to detect abnormal EV functioning by looking into vehicular operational information.

The individual contribution encompasses carrying out a thorough review of the literature on EV anomaly detection, researching appropriate parameters for operation, testing possible machine learning alternatives, and coming up with a solution for detecting deviations in the functioning of vehicles. As for the guidelines for selection of parameters, the study includes battery voltage, battery current, battery temperature, engine temperature, speed of the car, acceleration, power consumption, SoC (State-of-charge) and battery status depending upon their presence in the selected dataset.

Another important aspect of the contribution is to process the operational dataset so that it can be later analyzed. This includes analysis of the available data, solving issues related to quality of available data, selecting features and preparing the input data for development of the model and its testing. After that, the

appropriate anomaly detection method is applied to discover potential deviations in the behavior of the EVs.

The person's contribution covers the incorporation of the developed model into the SmartEV web application by way of back-end services. The incorporation allows for the processing of the existing EV data by the anomaly detection model and transmitting the obtained result to the front-end dashboard. The dashboard is aimed at sharing essential vehicle information and abnormally detected data for making users aware of the possible abnormal operating state of the vehicle.
What's more, the contribution also encompasses testing of the anomaly detection model with the help of the appropriate performance indicators and conducting the testing of the whole system at the level of the functioning of the integrated model and the dashboard.

1.9 Thesis Structure

- Chapter 1 – Introduction In this chapter, SmartEV, which is Electric Vehicle Management System using AI and ML. It explains the working of individual AI-Based EV Anomaly Detection System. It covers the research background, problem statement, research gap, aims and objectives, research questions and hypotheses, research scope, limitations, assumptions and contributions.
- Chapter 2 – Literature Review This chapter reviews previous research on electric vehicle monitoring, anomaly detection using machine learning, battery monitoring, fault detection in electrical and thermal sectors, motor abnormalities and charging issues. The chapter talks about different anomaly detection methods and machine learning techniques, data sources, and assessment techniques.
- Chapter 3 – Research Methodology This chapter outlines the research methodology that was used for the development and testing of the AI-Based EV Anomaly Detection System. The chapter mentions the research methodology adopted, the choice of data sets, the method of data collection, data processing, feature extraction, method of model selection, model creation, and method of performance assessment. The chapter includes further discussion on the evaluation of the performance of the selected machine learning method with the help of relevant performance measures.
- Chapter 4 – Product and Solution Design The chapter outlines the design of the proposed AI-Based EV Anomaly Detection System. This includes the system architecture in the general sense, the system design and development methods, integrated data flow, the ML workflow, and the dashboard design. The relevant architectural diagrams have been included in the chapter.

- Chapter 5 – Implementation, Integration and Testing This chapter elucidates the implementation process for each individual research component. It includes details on the technology utilized, dataset preprocessing, machine learning model development, implementation of back-end, integration with front-end, and dashboard development. Moreover, it gives a brief overview of the integration between the developed anomaly detection model and SmartEV web application along with the testing procedure carried out to validate the proper functioning of the developed system.
- Chapter 6 – Results and Validation This chapter contains the experimental results obtained during the research carried out with the developed anomaly detection model and the functioning system. Furthermore, it assesses the quality of the algorithm through the use of respective metrics, such as precision, recall, and F1-score depending on the type of detection used. In addition, the chapter gives information regarding the testing of the developed system and determines if the objectives of the research have been accomplished. All the results presented in the chapter are based on actual experiments.
- Chapter 7 – Discussion This chapter provides a discussion on the results regarding the research objectives, research questions, and the outcomes of previous studies. It also evaluates the effectiveness of the approach being used, the ability of the system to detect abnormal operating conditions, and the usefulness of the dashboard in presenting the detection results. The chapter also discusses the limitations, challenges, and implications for the practical use of the system.
- Chapter 8 – Commercialization and Sustainability This chapter discusses the expected uses of the AI-Based EV Anomaly Detection System and its commercial viability. The potential users of the system include EV owners, fleet owners, vehicle repair companies, and service companies. The chapter discusses the benefits of anomaly detection as well as the implementation, scalability, maintenance, and future possibilities for the solution.
- Chapter 9 - Conclusions and Future Work This chapter revolves around a recap of the research alongside the success rate of achieving the objectives and aim of the research. This chapter outlines the contributions made to the SmartEV research project, results noted from the research findings, and the limitations of the proposed system. Possible future works can be the application of real-life EV data for tests, taking more parameters into account, improving the effectiveness of anomaly detection, developing diagnostic accounts, and testing different EV models within the framework of the proposed system.

2.LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

The current thesis literature review has been centered on the studies which have been conducted in the field of electric vehicles (EV) monitoring, the use of machine learning for the purpose of anomaly detection, battery condition monitoring, detection of electrical and thermal failures, motor performance evaluation and intelligent vehicle control systems. The aims of the study presented here is to find the openings for the use of Artificial Intelligence and ML methods for better detection of abnormal state of electric vehicles and efficient monitoring decisions concerning vehicles maintenance.

This review presents the analysis of the previous studies done in the domain of EV battery monitoring, fault detection through machine learning, predictive maintenance, abnormal state detection and vehicle performance. This kind of research is necessary because it allows understanding the way machine learning algorithms can be used to analyze data concerning operation of vehicles and reveal unconventional patterns in it.

The review explores the significance of monitoring the operational parameters of EVs due to the fact that the performance of the vehicle is influenced by the interaction of different electrical systems, condition of battery, operation of motor, and process of charging. Different parameters like battery voltage, current drawn, battery temperature, motor temperature, speed of vehicle, acceleration, power consumption, State of Charge (SoC), and charging status give useful information about the operational condition of a vehicle.

The review also discusses the use of machine learning approaches for anomaly detection and examines the applicability of such techniques for deviation from the standard operation of the EVs. The basis for development of AI-Based EV Anomaly Detection System is created, which is a part of the larger SmartEV – Electric Vehicle Management System Using AI and ML. This component specifically deals with short-term anomaly detection instead of predicting the battery's State of Health (SoH) in the long run.

The review also discusses how machine learning models have been integrated with backend systems and web dashboards, which allows operational data to be analyzed and results of anomalies detected to be accessible for users.

The literature review has the following components:

1. Use of AI and ML in operating electric vehicles. Current methods of detecting and identifying faults in electric vehicles.
2. Electric vehicle battery monitoring and operational performance study.
3. Abnormalities that are related to electric, thermal, motor, and charging activities.
4. Operation parameters of electric vehicles.

5. Machine learning algorithms that are suitable for anomaly detection.
6. Preparations that are required for data selection and preprocessing.
7. Methods of measuring efficiency of machine learning methods in detecting anomalies.
8. Integration of anomaly detection models with the web-based monitoring systems.
9. Shortcomings of the current approaches in electric vehicle monitoring and gap in research that has been explored through this project.

The scope of this review is limited to published research and technical literature related to the design of a machine-learning-based EV anomaly detection component. It aims at identifying suitable operational parameters for data analysis techniques, anomaly-detection techniques, and approaches to system integration for identifying abnormal operational conditions in electric vehicles.

2.2 Domain and Theoretical Foundations

2.2.1 Electric Vehicle Monitoring and Management

EV monitoring is critical in the management of EVs, since the proper functioning of any EV depends on the performance of different interconnected components. These components include the battery pack, electric motor, power electronics, cooling system, and charging system. Changes in the performance characteristics of each of these components can have an impact on the performance of an EV.

In standard monitoring, the following methods are being employed. Predefined limits and vehicle diagnostics are used to detect certain problems. The limitations of these methods are that they are effective only in case of known faults. Finally, abnormal conditions can occur without a single parameter exceeding a predefined limit.

By utilizing machine learning, it allows for making improvements in monitoring electric vehicles through data analysis and detection of signs of normal and abnormal behavior. An advanced monitoring system can supervise different indicators and highlight deviations that would need closer inspection.

In particular, a research component is aimed at the creation of the AI-based electric vehicle anomaly detection system that, based on data available, detects possible abnormalities and unusual vehicle status. The goal is to use the technology in combination with traditional monitoring schemes and offer valuable data to the SmartEV web application.

2.2.2 Artificial Intelligence and Machine Learning in Electric Vehicles

Artificial Intelligence is defined as the set of techniques that allow systems to accomplish tasks such as pattern recognition and decision support. Machine Learning is the branch of AI that enables machines to learn patterns from data and exploit them to produce predictions or recognize different observations.
In connection to applications involving electric vehicles, machine learning can be applied to the problems of battery condition monitoring, estimating energy consumption, driving distance evaluation, fault finding, predictive maintenance, and anomaly detection.
Various techniques of machine learning can be used depending on the type of problem to be solved. If labeled data are available, supervised learning can be used to classify the data properly. Unsupervised learning allows identification of unusual events regardless of the labels. Finally, semi-supervised approach can characterize the normal operating conditions and judge whether there are instances of deviations.
In regards to the part of the project in question, the choice of the most suitable technique in machine learning would depend on the features of the available dataset, if anomalous events have been noted, and what results are expected from the detection. The method chosen is aimed to study the various parameters in use and to detect instances of abnormal behavior.

2.2.3 Electric Vehicle Operational Anomaly Detection

The method of anomalies detection consists of defining occurrences or trends that sharply contrast usual behavior. Observation of anomalies in electric vehicles means that data from regulators contradicts probabilities of standard values, the amount of heat generated from the engine exceeds norm limitations, unjustified consumption happens, engine operation is not regular, or charging behavior is abnormal.
Methods of anomaly detection can be grouped into techniques of statistical nature, rule-based techniques, supervised methods, unsupervised ones, and deep learning methods.
Statistical methods reveal the fact of deviation from the standard distribution or standard statistical threshold. Rule-based methods give parameters defined beforehand to define if a measure is abnormal or not. Techniques of supervised nature learn from pre-defined cases of normal and abnormal behavior while unsupervised methods are employed to find aberrations in the input data structure.

For instance, the unusual combination of the temperature in batteries, the current, and the power usage parameters may point to some checked operational mode. In a similar vein the extraordinary voltage changes or abnormal charging processes can illustrate certain anomalies.

Nevertheless, detecting an anomaly does not mean that we have either the proof of the existence of a separate malfunction or the reasons for it. Thus, the system should differentiate between the case of unusual observation and a case of previously registered failure. Thus, the recommended device will be responsible for the identification of possibly abnormal work modes for the diagnostics purposes.

2.2.4 Battery and Electrical System Monitoring

The battery plays one of the most important roles in electric vehicles. It must be monitored in order to correctly assess how the vehicle works. The battery voltage determines the electrical potential difference, whereas the current describes the flow of electricity in a battery circuit. These quantities change according to battery load, the amount and speed of charging, and other circumstances. Changes in voltage or current can signal possible malfunctions of the battery, although their interpretation depends on aspects such as the battery type, temperature, number of charging cycles, etc.

From the standpoint of battery performance, efficiency, and safety, battery temperature is a critical parameter because temperature relates to the performance. Any sudden increase in temperature may necessitate investigation, especially if there is unusual current or voltage.

The latest AI-Based Anomaly Detection for EV provides battery voltage, current, and battery temperature as the relevant input features whenever these data exist. Using operational parameters to link with their mutual relationship to allow for the possible identification of abnormal behavior that may not present itself when looked at in isolation. Detection results should be viewed against vehicle operational conditions and should not be construed as conclusive evidence of battery failure.

2.5 Motor, Thermal, and Charging-Related Parameters

Motor temperature should, along with the battery measurements, be considered for electric vehicle monitoring. It allows monitoring of the thermal state of the propulsion system. Abnormal temperature behavior can be caused by the load, the cooling system efficiency and other reasons but finding the root cause requires additional information obtained through special diagnostics.

Vehicle speed and acceleration can describe driving patterns and movements. Power consumption informs the electrical energy used by the vehicle depending on operational parameters. The evaluation of these measures in tandem can yield operating patterns that deviate from the expected ones.

Charging status also matters because the anticipated voltage, current, and power patterns differ while driving, charging, and being idle. Hence, the anomaly detection model must know the context of measurements. This component takes these parameters into account provided they are available in the selected dataset and contributes to the comprehensive analysis of the EV operating conditions.

2.2.6 Machine Learning Algorithms for Anomaly Detection

Various machine learning algorithms are applicable in identifying anomalous operations of electric vehicles. The appropriateness of these methods depends on their dataset properties, complexity of the operating phenomenon, availability of labelled anomalies and computational requirements.
Some of the most common methods include:

- Isolation Forest: This technique is used for the unsupervised detection of anomalies. This algorithm isolates cases using random divisions. Cases that are easier to separate are possibly anomalies.
- One-Class Support Vector Machine (One-Class SVM): The algorithm works by learning a border around normal cases, then identifying abnormal cases.
- Local Outlier Factor (LOF): This algorithm is based on density and detects cases that are much less dense than their neighbors.
- Random Forest and other supervised classifiers: All of these classifiers are able to separate normal and abnormal conditions when there is enough representative data.
- Autoencoders: These models are called neural networks that learn to rebuild input data. In case of large error in reconstruction, it is possible to detect cases that are inconsistent with those learned patterns.

There are advantages and disadvantages to every approach. Isolation Forest is effective for data sets with multiple dimensions, while the One-Class SVM method might be sensitive to both the choice of parameters and the importance of features. As for LOF, it is able to detect phenomena related to local deviations but is likely to depend on proper adjustment of neighborhood size. Supervised learning methods rely heavily on the availability and accuracy of the set of labelled training data, while autoencoders can only be employed on bigger datasets. Therefore, for the suggested system, the target algorithm has to be chosen based on experimental results from the existing EV data.

2.2.7 EV Operational Data Preparation and Feature Selection

Data preparation is a critical step in creating an anomaly detection system based on machine learning. Actual datasets may have missing values, duplicate entries, inconsistent measurement units, noisy data

obtained from sensors, and various parameters gathered at different time intervals. Data preprocessing consists of dealing with missing values, removing duplicates, monitoring measurement limits, standardizing measurement units, and analyzing unexpected values. The above methods must be used with caution since deleting unexpected values without appropriate analysis may also destroy real anomalies that the system is designed to find. Feature selection entails selecting the operational parameters that convey information useful for anomaly detection. For the proposed device, suitable parameters may be voltage, current, battery temperature, motor temperature, etc.

One should also consider the need for feature scaling as some algorithms are more sensitive to features measured in their numeric ranges. For instance, it is known that voltage and temperature metrics are expressed on different scales while the charging status may only appear as a nominal variable. The data preparation procedure must be designed according to the data set and algorithm selected. The training and testing data must be separated properly in order to avoid data leaks and perform a more accurate evaluation.

2.2.8 Evaluation of Machine Learning-Based Anomaly Detection

It is important to assess the performance of anomaly detection models to ascertain their effectiveness in detecting anomalies. Which evaluation metric to use depends on the type of dataset, specifically whether it contains labeled anomalies, and the prediction nature of the anomaly detection model. When both positive and negative labeled samples are available, confusion matrices can be employed in order to compute true positives, false positives, true negatives, and false negatives.

Some of the commonly used evaluation metrics are:

- Precision is the ratio of the number of observations identified as anomalous which are actually anomalous according to the reference labels available.
- Recall is the ratio of the number of known anomalous observations identified by the model correctly.
- F1-score is the harmonic mean of precision and recall.
- Accuracy is the ratio of all correctly classified observations when there is a suitable labeling classification scheme available.
- ROC-AUC is an effectiveness measure at all the decision thresholds when reference labels and the prediction scores are available.

It is possible for accuracy to show that a certain observation is true. As an example of this, accuracy may turn out to be high due to the model predicting the majority class. However, precision and recall should be

used together when labeled data are available. In the case of unsupervised methods, evaluation can use labeled test sets, observations evaluated by an expert, or designed evaluation systems. However, if there are no credible labels for the observations, one should state this limitation in the evaluation. The present work will follow the needed evaluation methodology and evaluation of the obtained results will be based on the characteristics of the data set which is used in the analysis.

2.2.9 Web-Based Decision-Support Systems

Decision support systems make the users accessible to information required for monitoring, assessing, and taking decisions of the actions taken. All DRS have the frontend which refers to the interface and all backend services in addition to data storage and means of doing analytical operations. For Machine learning-based EV monitoring application, the backend will receive driving data from vehicles, verify the information as input, run anomaly detection, and get outputs to the frontend. The information at the frontend is presented as vehicle metrics, indicators of anomalies, and other data which make it possible to delve into the process of investigating why driving was abnormal. The SmartEV application is considered to be the general platform for the component of anomaly detection. The dashboard serves the purpose of making the results of the model visible to the users and performing investigations into any technical issues that can occur in the process. It is important that technology description coincides with the technology being used. In an exact way, React.js, Node.js, Express.js, and Firebase or MongoDB is to be indicated only after they are used in reality.

2.2.10 Integration of EV Operational Monitoring and Machine Learning

The combination of EV operational monitoring along with machine learning acts as a method to convert raw vehicle data into valuable results in terms of anomaly detection. The first step in the process consists of gathering operational data from the selected data source, while the second step consists of preparing the data for further processing by the trained model. The third component of the process consists of passing the data to the trained machine learning model. Once this is done, the model will analyze the data given to it and produce the anomaly score, anomaly label, or any other output based on the method it uses. The result of this process can be delivered by backend services and represented on the dashboard.
Thus, the whole process described can be depicted in the following flow.

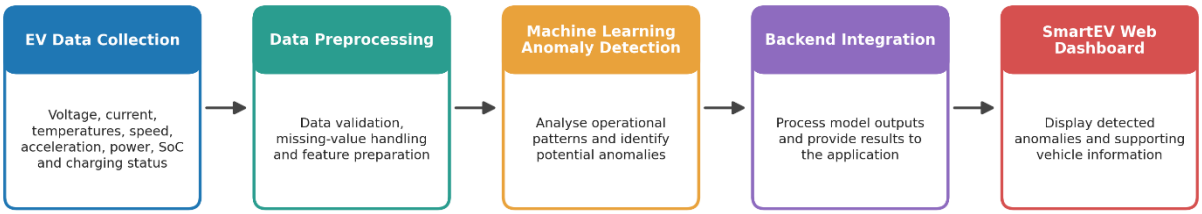


Figure 1 AI-Based EV Anomaly Detection Workflow

3 RESEARCH METHODOLOG

3.1 Research Design

The research uses its design and development research methodology in order to come up with the AI-Based EV Anomaly Detection System, which is part of the Electric Vehicle Management System Using AI and ML called SmartEV. This element of the system is intended to detect extreme vehicle operations. The aim of the study is to devise an anomaly detection framework based on data got from the operation of EVs through the analysis of parameters such as battery voltage, current, battery temperature, motor temperature, vehicle speed, rates of acceleration, power consumption, State of Charge (SoC), and charging status.

The research comprises of articulating the research problem statement, acquiring and preparing a relevant dataset, creating a suitable machine learning model, evaluating its performance, and incorporating the detection part into the SmartEV project. Figure 3.1 outlines the research activities and how they relate to the dissertation.

Table 1 Research activities and their relationship to the study

Research activity	**Activity in this study**	**Thesis chapter**
Identify the problem	Identifying problems related to abnormal operation of electric vehicles	1, 2
Define the problem	Creation of the AI-based system for the detection of electric vehicle anomaly	1
Literature Review	Review of different topics related to electric vehicle monitoring, anomaly detection, and machine learning	2
Design and development of the system	Preparation of a dataset, feature extraction, and construction of an anomaly detection model	3,4,5
Demonstration	Demonstrating images of the found anomalies through the interface of SmartEV.	5,6
Evaluation	Using appropriate metrics for assessing the developed model	3,6
Communication	Writing the thesis, reports, and research papers	All

As shown in Table 3.1, the following activities represent the basic stages of the development of the AI-Based EV Anomaly Detection System. These stages include reviewing the literature, preparing EV operating data, development of the machine learning model, model integration, and result documentation.

3.5 Data, Participants or Study Materials

3.2.1 EV Operational Dataset

The main material in this research is a dataset of electric vehicle operations that provides data about the operation condition of the vehicle and the performance of the vehicle. This dataset is utilized for the development and testing of an anomaly detection model belonging to SmartEV. The model analyzes all EV parameters such as the voltage of the battery, current, the temperature of the battery, the temperature of the motor, vehicle speed, acceleration, power consumption, State of Charge (SoC), and the charging status. By analyzing these parameters, the model identifies abnormal patterns indicating irregular vehicle operations. The dataset is important for model building and testing. Once the model is built, it can be included in the SmartEV application and provide the users with information on the detected anomalies in the operation of vehicles.

Table 2 Main study materials

Study material	**Purpose**
Dataset of electric vehicles	while learning as well as analyzing the anomaly detection process
Electric battery's voltage and amperage	Detection of unusual current flow patterns.
Temperature readings of motor and battery	Deduction of abnormal heating.
Speed and acceleration of the vehicle	Study of abnormal operation of the vehicle.
Energy consumption	Spotting unusual consumption of energy
State of Charge	Delivering battery charging information
Charging condition	Studying the operating conditions related to charging
Machine learning algorithm	Detecting abnormal patterns in electric vehicle performance

SmartEV	Showing results of anomaly detection
Back-end environment and storage	Implementing the architecture for data processing and result output

3.2.2 Study Participants

A key aspect of the main model development process is based on the operational electric vehicle data rather than obtaining first-hand information via surveys and questionnaires. This means that the focus of the model development experiment will be on analyzing the available data with the purpose of finding abnormal patterns in the vehicle operation process. The component that is developed aims to help SmartEV users by providing them with information about possible electric vehicle abnormalities. The users will be able to see the results of the identified patterns through the application. Whether there will be any further evaluation with the help of electric vehicle owners, drivers, or specialists, it should be done separately referring to model training.

3.3 Experimental or Study Procedure

The process of the research includes these key phases:

- Choosing and preparing electric vehicle operational data.
- Deciding on the suitable electric vehicle operation features.
- Development of features and data preparation.
- Creation of machine learning model.
- Evaluation of the model and its efficiency.
- Implementation of the detecting model in SmartEV.
- Forecasting of abnormal situations and result producing.
- Demonstration of detecting outcomes by means of the app.

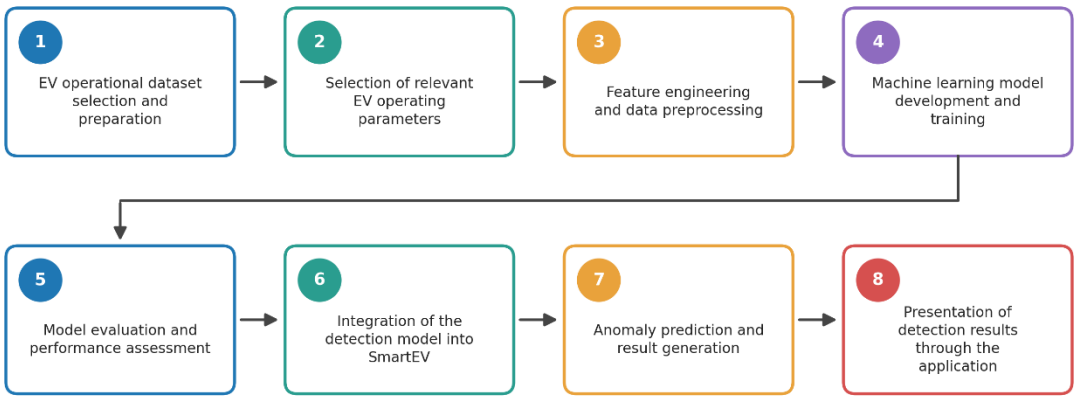


Figure 2 Experimental Procedure for the AI-Based EV Anomaly Detection System

3.3.1 Dataset Preparation

This stage will transform the selected electric vehicle operational dataset into a format suitable for the development of the machine learning model. The dataset is therefore accessed for missing values, duplicate records, inconsistencies in formatting, and invalid measurements-all factors that might impair the performance of the model. Relevant parameters such as battery voltage, current, temperature, vehicle speed, acceleration, power consumption, and state of charge (measured the SoC) are selected based on their availability in the dataset. Appropriate preprocessing techniques are applied to prepare the data for training and testing. Thus, the cleaned data is examined that paves the path for the detection of normal and abnormal electric vehicle operation modes. Quality and suitability of the dataset are of computational relevance, as reliability and robustness of a machine-learning model depend highly on those inputs.

3.3.2 Model Development and Training

The dataset that has been created plays a crucial role in building a machine learning model that can be trained on it for detecting anomalies in vehicle operation. By assessing the selected operational parameters, the model can find patterns that do not match the expected vehicle operation. The detection process aims to spot abnormalities in terms of electrical performance characteristics, temperature levels, motor operation data, and energy consumption in addition to checking charging conditions. Depending on the chosen algorithm and labels the dataset will have, the system will be able to detect abnormal patterns or categorize the state of operations. The trained model is expected to become the intelligent part of the

SmartEV solution. After the integration is completed, the model will start working with operational data and use it to make predictions presented through the application.

3.3.3 Model Evaluation

This paragraph presents the evaluation process of a machine learning model that was created for identifying unusual operating conditions in electric vehicles. The evaluation process employs some dataset and testing strategy to assess the model's performance. Depending on the choice of the model and the available labelled data, the performance can be estimated based on the effectiveness of the model given by some performance metrics like accuracy, precision, recall, F1-score, or using confusion matrix. Based on the performance assessment, the weaknesses and strengths of the model can be revealed. Specifically, the false alarms and missed anomalies are important because they can limit the ability of the detection system to fulfill its purpose. After the model has been evaluated, it can be used along with SmartEV app for processing the data on EVs and getting the predictions.

3.3.4 EV Data Collection and Communication

The AI-Based EV Anomaly Detection System is dependent on operational information in order to reflect the instant status of a given electric vehicle. Data may come from chosen datasets or come through data processing functions used within the SmartEV app. The system processes given values and passes them on in the format expected by the anomaly detection system. If the system architecture allows, the backend is responsible for ensuring the data processing and relaying the output information to the frontend. Results of the analysis are shown in the app interface to let users understand abnormal situations. It is evident that machine learning analysis can be integrated into the general work of the SmartEV system. The focus of the component is made on software-based analysis of EV operational data. Any hardware solution for data collection or communication should be mentioned only if they are part of the final version.

3.3.5 Feature Extraction and EV Anomaly Detection

Anomaly detection is a process driven by the operational parameters available in conjunction with a selected EV dataset. These may include battery voltage, current, battery temperature, motor temperature, vehicle speed, acceleration, power consumption, SoC, and charging status. Once the prediction is made, input data is checked and converted, if necessary, to meet the characteristics and format preconditioned by the trained model. Feature selection and engineering identify relevant relationships among the operating parameters to improve the consistency of the detection process. Thus, a trained ML model examines input features to identify operating patterns that are actually outside the defined expected behavior. For instance, anomalies that may flag further investigation could include abnormal temperature readings, unexpected bridge variations, or other unusual power consumption when bolstered by both the

model and dataset. Lastly, the outputs from the model shall be made available on the SmartEV application so that the users may be warned about the suggested EV abnormalities. Besides, the results from the model may support preliminary diagnosis and further technical investigations. Yet, the anomaly prediction alone is not sufficient evidence of a definite mechanical or electrical fault, hence further inspection may need to be performed.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Stakeholders

People involved with the AI-Based Electric Vehicle Anomaly Detection System include people who are affected by the system or its outcome, so that they understand operating conditions in the electric vehicle (EV). The system collects input data from the electric vehicles in order to discover anomalies and to give information regarding their occurrence.

Table 3 Stakeholders and Their Roles and Interests

Stakeholder	**Role and interest**
EV Owner	Reviews vehicle condition information and receives alerts about potential abnormalities
EV Technician	Uses anomaly detection results to support further vehicle inspection and troubleshooting
System User	Accesses the application and reviews available EV monitoring information.
System Administrator	Manages system access, application configuration and technical maintenance
Developer	Develops, evaluates and maintains the machine learning models and its integration with SmartEV

4.1.2 Functional Requirements

The functional specifications are an account of the principal activities of the AI-Based EV Anomaly Detection System. This covers receiving the operational EV data, detecting unusual patterns, forecasting, and displaying the outcomes on the SmartEV application.

Table 4 Functional Requirement

Functional Requirement	**Description**
Data Input for Electric Vehicle (EV)	The electric vehicle system should accept the available operational parameters related to the EV.
Validation of Data	The electric vehicle system should validate the input data before processing it.

Data Preprocessing	The electric vehicle system should process the operational data and prepare it for the model as required. Selection of Features The electric vehicle system will only use the relevant features from the EV such as voltage, current, temperature, speed and the state of charge if there are any available.
Detection of Anomalies	The electric vehicle system should detect the existing abnormal operating conditions of EV through the use of a chosen machine learning model.
Classification of Anomalies	The electric vehicle system should classify the existing anomalies according to the operation of the chosen model.
Report Creation	Anomaly detection reports should be produced by the system for the input data provided.
Report Display	The implementation of the detection reports should be done by the system using the SmartEV app.
Old Report Storage	Whenever feasible, the system should save the reports so that they can be viewed later.
Diagnostic Assistance	Information that aids users in identifying possible problems with electric vehicles must be given by the system.
Mistakes Management	The system should correctly manage mistakes and errors connected to prediction.
Integration of Models	By means of the integration system the trained model should be connected with the application.

4.1.3 Non-Functional Requirements

Non-functional requirements describe expectations for the quality features characterizing the anomaly detection system, including efficiency, dependability, ease of use, safety, etc.

Table 5 Non-Functional Requirements

Quality Attribute	**Description**
Efficiency	The given system is required to work with available data on EVs and emit forecasts within an acceptable time.
Dependability	he given system should be capable of properly dealing with the absence and un/partially incorrect data.
Ease of use	The given interface should clearly demonstrate the results of the system.
Safety	Access to vehicle data and system operations should be limited if the authentication is employed.
Integrity	The operational history and forecasts of EVs should be recorded without inconsistencies.
Maintainability	The system should follow a modular design enabling ease of troubleshooting.
Scalability	The architecture should be able to consider an increased number of vehicles and more operation records in the future.
Confidentiality	Vehicles and users' information should be treated properly.

4.1.4 User Scenarios

User scenarios give details about the processes through which the users of the AI-Based EV Anomaly Detection System will be interacting with the system. The primary scenario entails the submission of EV operational data, the validation of that data, the processing of EV operational data through the trained model, and the display of the detection results.

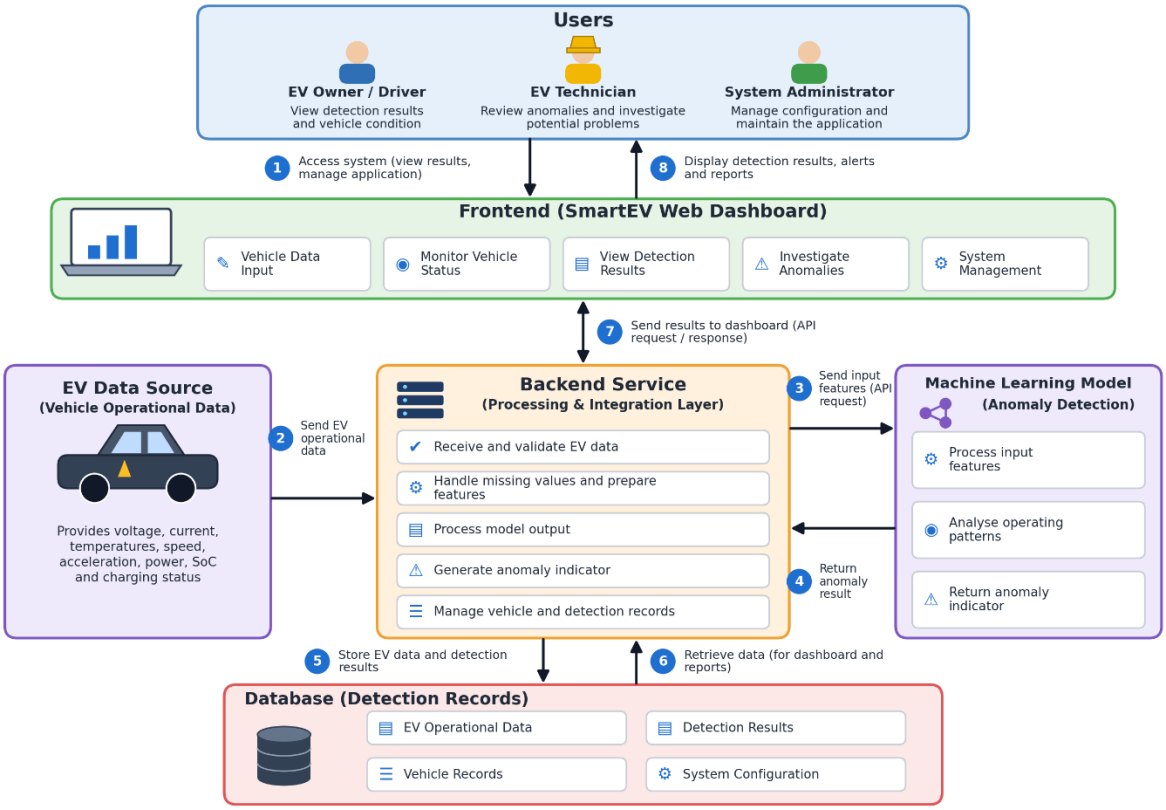


Figure 3 User Scenario for the AI-Based EV Anomaly Detection System

Table 6 User Scenarios

User Scenario	**Primary Actor**	**Expected Outcome**
Provide EV Operational Data	System user/Data source	Relevant vehicle parameters are provided for analysis.
Validate Input Data	System	Incorrect or incomplete values are processed.
Process EV Data	System	The input is amended for the purposes of model processing.
Identify Irregularities	Machine Learning Model	Tool Exceptional process behaviors are discovered.
Obtain Detection Information	System	A forecast or anomaly notice is generated.

Observe Detection Information	EV Owner / Driver	The indivi dual can see the detected data.
Analyze Suspicious Issues	EV Engineer	The output enables additional mechanical investigation.
Record Detection Data	System	The output is recorded if durability is ensured by the development.
Control the App	System Administrator	The system is configured and maintained.

4.2 Component Architecture Within the Complete Solution

The Electric Vehicle Management System using AI and ML is called SmartEV which is designed to integrate EV management operations together with smart analysis. The AI-based EV Anomaly Detection System is a separate module that analyses the parameters of EV operation in order to detect any abnormal activities. The specific component can be structured by means of multilayered software architecture including frontend, backend or integration service, data storage and the machine learning component. The specific technologies to be used depend on the implementation of SmartEV.

Key elements of architecture:

4 Frontend: It offers an interface for users to enter EV data or view any detected abnormalities.

5 Backend/API: This component is responsible for data validation, conducting processes for the application, and sharing predictions.

6 Data storage: It includes storage of relevant information about the vehicle and its operation.

7 Machine learning model: It analyses the input operational parameters to detect anomalies.

8 Anomaly detection results: These results give information on the detected anomalies.

9 SmartEV integration: It connects the anomaly detection unit to the bigger EV management application.

Operational data is input into the system and is validated prior to processing. The relevant features are routed to the machine learning model that has been pre-trained. This model produces output with location determinations that are returned to the application and subsequently shown to the user. In case of historical storage implementation, the results are also saved for future use.

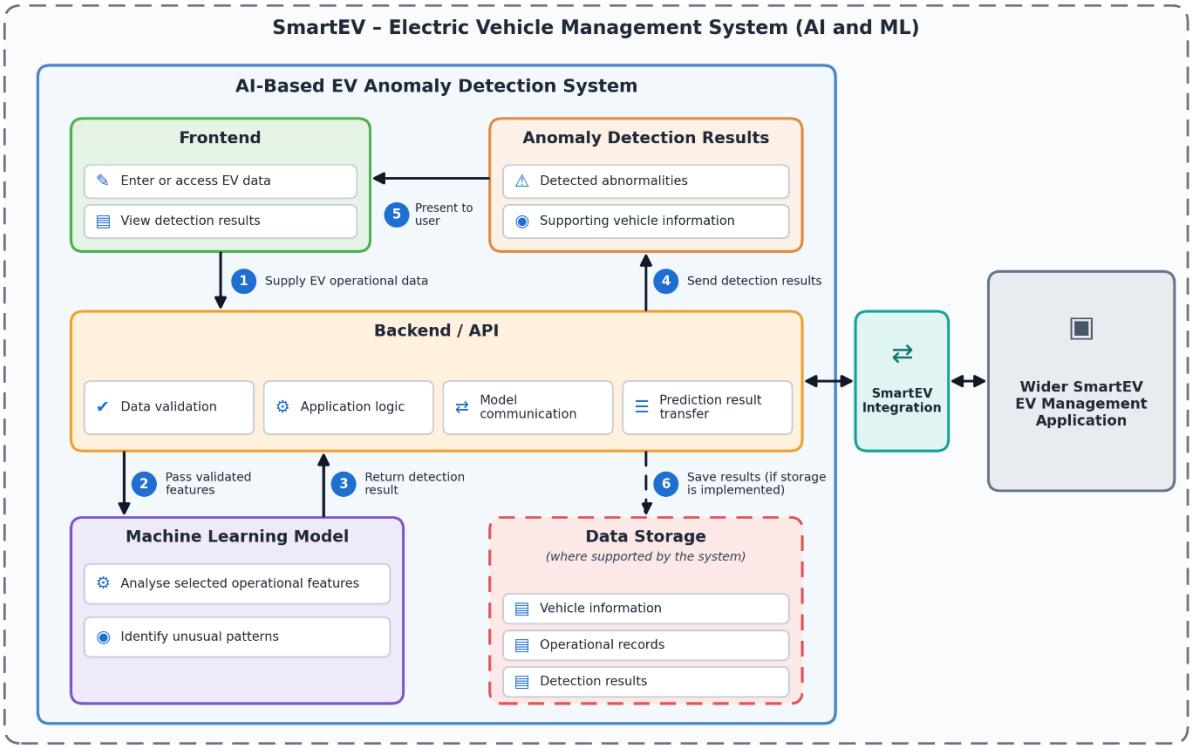


Figure 4 Component Architecture of the AI-Based EV Anomaly Detection System

4.3 Design Alternatives and Decisions

There are a number of design considerations to keep in mind when developing the AI-Based EV Anomaly Detection System. Generally speaking, these include machine learning algorithms, application architecture, data management, feature extraction and results presentation.

Table 7 Design Alternatives and Decisions

Design Element	Alternatives	Design Decision	Reason
Machine Learning Algorithm	Classification, clustering, or outlier detection methods	Pick an algorithm appropriate for data at hand	The system must be capable of identifying unusual electric vehicle operational patterns.
Input Parameters	Single variable or multiple functioning variables	Utilize only appropriate available features of EVs	Use of several parameters will allow a better understanding of

			the vehicle operation.
Data Preparation	Original data or cleaned data	Perform data validation and preparation prior to predictions	Improves consistency of the input and ensures reliability of processing.
Frontend	Various web or app frameworks	Use framework chosen for SmartEV	Facilitates incorporation into the project as a whole.
Backend	Frontend only processing of processing with the help of the backend	Apply architecture that can be implemented	Helps separate logical reasoning and model from one another.
Data Storage	Systems that use either relational or document-based model, or yet no persistent storage	Use the SmartEV data architecture chosen.	Allows adequate management of the records related to vehicles and prediction.
Result Presentation	Only raw data or an error signal and its description	Present user-friendly results.	Assists users in understanding the results produced by the algorithm.
Error Handing	Simply disregard the incorrect data or validate it prior to processing	Validate the input as well as handle prediction errors.	Ensures that improper input does not go unverified.
Model Evaluation	Use informal observations or numerical evaluation	Evaluate the model using appropriate metric.	Stablishes how well the model works.

The proposed system is mainly based on the machine learning model which is determined based on the properties of the dataset at hand and on whether or not there are any identified anomaly records in it. The method chosen must be tested prior to its use in the application. The frontend and backend portions must make it possible to transmit the data from EVs and demonstrate the predictions made. The approach used for the storage of the data must be the same as that applied in SmartEV, while preprocessing and input validation tackle consistency issues. In essence, the system must demonstrate anomalies as unusual conditions instead of confirmed vehicle faults to guide the user on whether there is a need for the further investigation.

4.4 Detailed Design

The detailed design of the AI-Based Electric Vehicle Anomaly Detection System describes how operational information, preprocessing, the machine-learning model, the application, and data storage cooperate with each other. Main stages consist of getting vehicle input, validation of input, feature preparation, anomaly detection, result generation, and result presentation through SmartEV.

4.4.1 Frontend Design

The user interface has already been designed in advance for the interaction with the anomaly detection module. It serves as a medium to present available EV data and also presents the result acquired by the machine learning model. The main functions of the frontend are as listed below:

- Provide information for the available operational parameters of the EV.
- Present the results of the anomalies detected.
- Provide information on the condition of the vehicle.
- Present the anomalies in a form that is easy to understand.
- Access to the records of previous detection of the anomalies.
- Provide necessary information when the input data are corrupted or prediction cannot be performed.

The interface needs to make the difference between the standard information provided and the one where the anomalies are identified based on the models. The easy designation of the labels helps the user to understand the results without supposing knowledge in machine learning.

4.4.2 Backend Design

The backend or integration service facilitates communication between the various components, such as the application, operational data, machine learning model, and data storage where it might be applicable. The principal duties of the backend are validating the incoming data, formatting it properly, sending the data to the model, getting the model outputs, and sending the answers back to the frontend. The backend should also be able to maintain the database of the predictions and deal with possible errors occurring in the process. For example, if certain parameters for the input are either not present or incorrect, then the system should provide the correct statement rather than making the ungrounded prediction. Thus, the requirements for the backend architecture and technologies need to adhere to the real architecture utilized in SmartEV.

4.4.3 EV Data Collection and Communication Design

The anomaly detection module demands the operational statistics on an electric vehicle. Depending on the data set and the realization of the system, such parameters might include the battery voltage, battery

current, battery temperature, motor temperature, speed of the vehicle, acceleration, power consumption, State of Charge (SoC), and charging state. The submitted data can either come from one of the existing databases or from the implemented vehicle-data interface. Before making predictions, the system verifies the given values and brings them to the form which is required by the model. The module handles EV operational data. Only if a direct link to the vehicle sensors, onboard systems or IoT devices is used in the implementation should it be included in the design.

4.4.4 Machine Learning Model Design

The machine learning module is designed to detect any abnormal patterns when an EV is being operated. The features of the operational data that are received from the preprocessed data are used in the module to apply an anomaly detection approach. An example of the input data includes power consumption, motor temperature, battery voltage, and vehicle acceleration. The input features that are used depend on both the dataset being used and the model at hand. In model development, dataset preparation, feature selection, fitting the model selected, and assessing performance must be conducted. In predictions with the model, input data must be identified in the same feature structure and preprocessing steps used when the model was developed. Depending on the type of algorithm selected for receiving results, the nature of the output may vary as it can either say if the observation is anomalous or it can provide a certain diagnosis of the condition. Before using the model results, the evaluation of the model must be conducted.

4.4.5 Database Design

The design of the data storage system enables the management of the data required for the operation of the AI Based EV Anomaly Detection System. The database design differs based on the data storage technology and the design of the database schema in use in the SmartEV system. The most important types of information that could be stored are:

- Vehicle information: Vehicle identifiers and other important details of the vehicle.
- Operational data: EV readings necessary for the analysis.
- Processed features: Chosen or converted input features needing storage.
- Detection results: Model outputs, indications of anomalies and corresponding timestamps when available.
- Prediction history: Past detection information for later use, if supported.
- System records: Necessary application data needed to ensure functioning and maintenance of the system.

The application gets information from the database to display vehicle information and results of detection. Data validation and access rules are applied to provide data integrity and protect stored information.

4.4.6 Anomaly Detection Results and Dashboard Design

After processing the operating data, the machine learning algorithm presents the outcome of the detection task in a form approachable for end users. The dashboard could include the current vehicle parameters, anomaly detection information, necessary input data, and prior detection outputs where applicable. In case the detected anomaly is accompanied by the anomaly score or confidence level, such data could also be represented accompanied with its explanation. The outcome must assist the user in deciding on whether to proceed with further investigation in case of detection of a possible abnormality. However, the result cannot be interpreted as evidence of a certain vehicle defect before receiving confirmation from an independent entity. This implementation allows the AI-Based EV Anomaly Detection System to generate decision-support data in SmartEV while keeping the separation between predictions made by the algorithm and confirmed technical diagnosis.

4.5 Standards and Professional Practice

Both the designing and the development of the Artificial Intelligence Based Electric Vehicle Anomaly Detection System must be carried out according to different software engineering techniques, principles of responsible machine learning, and the rules of data protection.

- Data verification requires the confirming of parameters of operation as well as proper dealing with the situations when the required data is not available, invalid, or contradictory.
- Application security means protecting of the main application components as well as passwords and data placed in its memory.
- Regarding data protection, it is necessary to work with vehicle data as well as customers.
- Disclose the aspects of the use of anomaly recognition technologies.
- Evaluate models according to the measurement method that has been chosen.
- Keep in mind that it is necessary to manage licenses while working with different pieces of software.
- Maintainability of the project simply means that the system was developed in a modular way allowing to search for errors, testing and further development of the operation.
- Reference the sources according to IEEE format.
- Error handling should be organized according to the latest standards.
- Human oversight is meant to be an obligatory part of the process of the anomaly explanation.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The Intelligent Electric Vehicle Anomaly Detection System was created as a part of the Electric Vehicle Management System Using AI (SmartEV) to allow the integration of new technologies in order to process the Electric Vehicle data, to solve the task of abnormal operation detection within Electric Vehicles. The development of the component consists of collecting and preparing data, and developing a model.

- Front-end framework is utilized for showing data regarding electric vehicles and detection outcomes of abnormalities.
- Back-end framework is needed to process requests and communicate with machine learning model.
- Machine Learning Algorithm deals with finding abnormal patterns of electric vehicle usage.
- EV Operation Dataset is used for the purpose of getting the data for creating models and model tasks.
- Data Storage refers to acm storing information associated with vehicles and predictions.
- Google Colab is a tool for developing and training the model in cases when it is used in the implementation.
- Visual Studio Code is a tool for writing and managing code used to create the application.
- Postman is a tool for backend API testing to some extent.
- Git and GitHub are a tool for carrying out source code management and version control where it is used.

5.2 Component Implementation

The AI-Based EV Fault Detection System provides the capability for EV operation data processing coupled with machine learning-based anomaly detection. This component gets the list of non-constant EV parameters, and it does the necessary manipulations to make input data available for the trained model. The dataset can include battery voltage and current, temperature of battery and motor, vehicle speed and acceleration as well as power consumption and state of charge (SoC) and charging state; selected features depend on the type of used dataset. After receiving all the information, the system verifies the supplied information and gets it ready for appropriate processing steps. Finally, the system transmits the selected features to the machine learning model, which analyzes the performance of the system and finds potential abnormalities. Based on the structure of the model and the data used for its creation, it produces the anomaly detection result which is presented to the user via the interface of SmartEV application. This system can also arrange the storage of the anomalies for future use in cases they need to be analyzed.

5.3 Integration with the Other Components

The SmartEV project features the AI-Based EV Anomaly Detection System as part of the Electric Vehicle Management system leveraging advanced AI and ML techniques for electric vehicle management. Integration is a procedure that consists of connecting the anomaly detection model to the application services performing the operational data receiving, input processing, and output presenting. According to the architecture, the front-end provides the users with the detection results while the back-end or the integration service coordinates communication with the machine learning service. The module receives the available EV operational parameters, performs all necessary preprocessing and submits them to the model. Once processed by the model, the information about the detected anomaly is communicated back within the application. In accordance with the overall architecture of the system, the information about the detected anomaly will be used in conjunction with other EV management system information when applicable.

5.4 Verification and Software Testing

It is essential to evaluate AI-Based EV Anomaly Detection System through testing. It includes input validation, data preprocessing, feature engineering, model execution, producing results of anomaly detection, system integration, and reporting results. The treatment of test cases in Table 5.2 indicates how the component of the system operates. They have to be run on the functioning system and the results need to be documented before submitting the final thesis.

Table 8 Component Test Cases

Test ID	**Test Scenario**	**Expected Result**
TC01	Use valid operational data of EV	The system will accept the provided input data.
TC02	Provide missing or incomplete values	The system will recognize the missing information and address it accordingly.
TC03	Provide erroneous operational data	Evidence of invalid data will be removed or processed depending on the validation rules.
TC04	Transform the valid EV data	Data is converted in a required format for the analysis.

TC05	Select required features	Required features are processed by the model.
TC06	Feed the prepared data into the model	The model is able to provide an output.
TC07	Analyze data consisting of abnormal operating pattern	The model returns an anomaly outcome with respect to that the pattern fulfills its criteria for detection.
TC08	Process data depicting permissible operating behavior	The model yields the result as per its acquired techniques or defined decision rules.
TC09	Show results of anomaly detection	The solution is able to display the produced result accurately.
TC10	Save a detection record	The result is stored if the database storage has been employed.
TC11	Retrieve obsolete detection results	Present previous records correctly.
TC12	Manage the fault of the system or service	The application returns a proper error message without recognizing the prediction as a proper one.

The testing process must also assess model performance using the relevant metrics appropriate to the algorithm and dataset being used. Based on the availability of labelled data, it may involve metrics such as precision, recall, F1-score, accuracy and confusion matrix. Functional testing confirms whether or not the functionality behaves as expected while the evaluation of the model indicates its capacity to detect abnormal operational patterns. The report should consist of real testing results and performance values after testing has been performed.

5.5 Deployment and Operation

The AI-based EV anomaly detection system functions as a component of the SmartEV application. It is dependent on the availability of a required data source, preprocessing logic, machine learning models trained to date, application services as well as data storage, if applicable. During its operation, the system receives operational data from EVs and checks whether it is valid. The accepted data is processed

according to input requirements of models and fed into a trained model for anomaly detection. Then the model analyzes this input and provides a result indicating whether the anomaly is found or not. The obtained result is sent back to the application and displayed in a user-friendly way. If the historical storage is supported, the obtained result may be retained for future comparison purposes.

To ensure that the system works correctly, the operational configuration of application services, models, data preprocessing, credential management, and error handling must be done correctly. Moreover, the system must deal with missing data or failure of the predictive models without producing false conclusions. The effectiveness of the deployed solution largely depends on the quality and suitability of the data for the modeling process, the chosen models, and the conditions in which the results were obtained. Therefore, any accuracy and performance figures discussed in the paper must correspond to the conducted experiments.

6. RESULTS AND VALIDATION

6.1 Research Results

The purpose of my research is to create an AI-Based Electric Vehicle (EV) Anomaly Detection System (AIEVADS) which is capable of detecting abnormal vehicle operating conditions and provide a framework for predictive maintenance using Artificial Intelligence (AI) and Machine Learning (ML) algorithms, by analyzing the EV operational data and detecting patterns of abnormal behavior that may indicate potential mechanical or electrical faults.

The operational parameters which are considered for the research include battery voltage and current, battery, motor temperature, state of charge (SOC), vehicle speed and acceleration, charging status, and power consumption. The parameters will be used to create models in which machine learning algorithms that detect abnormal conditions in electric vehicles will be developed and evaluated.

Three machine learning models are under consideration for comparison: Isolation Forest, One-Class Support Vector Machine (One-Class SVM), and Deep Autoencoder Neural Network. Isolation Forest detects anomalies based on the isolation of unusual observations. One-Class SVM learns normal vehicle behavior patterns, distinguishing normal from anomalous observations. The Deep Autoencoder Neural Network learns the representation of normal operating data and detects anomalies based on reconstruction errors.

The comparison of the performance of the three machine learning models will be carried out by calculating relevant machine learning performance metrics, including Accuracy, Precision, Recall, F1-score, ROC-AUC, and prediction time if applicable. A comparison of the results will be made to select the model that best suits the use case in terms of performance and speed.

Table 9 Evaluation Metrics for EV Anomaly Detection Models

Evaluation Metric	**Description**
Accuracy	Measures the proportion of correctly classified observations when ground-truth labels are available.
Precision	Measures the proportion of predicted anomalies that are actual anomalies.
Recall	Measures the proportion of actual anomalies correctly detected by the model.
F1-score	Provides a combined measure of precision and recall.
ROC-AUC	Evaluates the ability to distinguish normal and anomalous observations when suitable labels and prediction scores are available.
Prediction Time	Measures the time required by the model to produce an anomaly detection result.

The final results of the evaluation of the Isolation Forest, One-Class SVM, and Deep Autoencoder will be determined once these models are trained and tested on the selected electric vehicle datasets. The comparison will highlight which algorithm is more suitable for the task proposed for the system.

The chosen model should be capable of recognizing abnormal vehicle operating conditions and detecting anomalies that can generate maintenance alerts, vehicle status reports, diagnostics reports about the vehicle's malfunction level, and recommended maintenance recommendations. All of these features will benefit EV owners or service stations by informing them of possible issues observed with their cars.

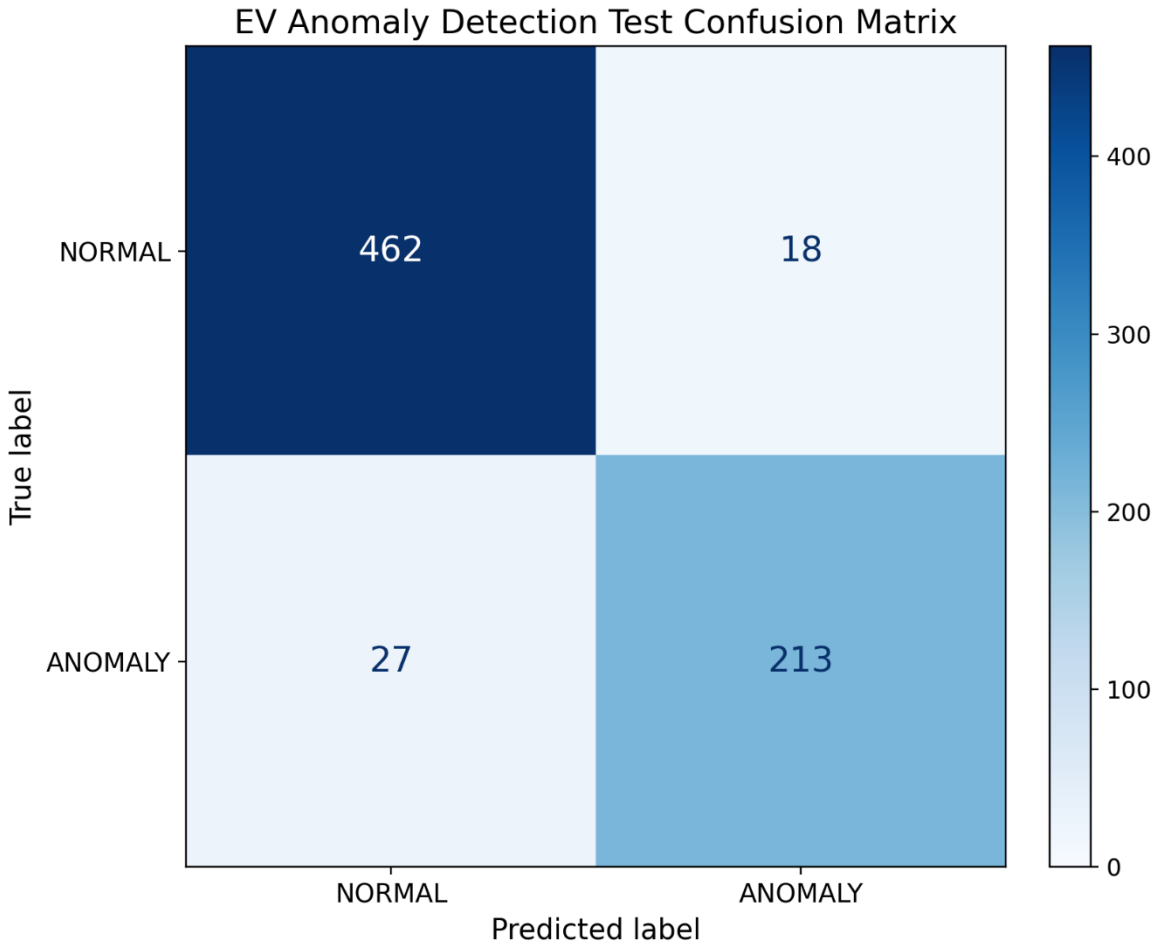


Figure 5 Comparative Evaluation of EV Anomaly Detection Models

```
Classification report:

              precision    recall  f1-score   support

      NORMAL     0.9448    0.9625    0.9536       480
     ANOMALY     0.9221    0.8875    0.9045       240

    accuracy                         0.9375       720
   macro avg     0.9334    0.9250    0.9290       720
weighted avg     0.9372    0.9375    0.9372       720
```

Figure 6 EV Classification Report

```
============================================================
EV ANOMALY DETECTION MODEL — TEST RESULTS
============================================================
Accuracy:          93.75%
Balanced Accuracy: 92.50%
Macro Precision:   0.9334
Macro Recall:      0.9250
Macro F1:          0.9290
Weighted F1:       0.9372
```

Figure 7 EV Test Results

6.2Product and Technical Results

The proposed AI-Based EV Anomaly Detection System will be an intelligent part of an Electric Vehicle Management System. The system will utilize machine learning models, Python back-end, a mobile application, and Firebase database for providing vehicle health and predictive maintenance services.

The system will process the operational vehicle data and provide information about the abnormal functioning of the vehicle. The proposed machine learning model will be hosted on the Python back-end

which will fetch the data from the mobile application and provide the results in the form of vehicle health information.

The main technical aspects of the proposed AI-Based EV Anomaly Detection System include:

- EV operational data processing: It will collect the battery voltage, battery current, battery temperature, motor temperature, vehicle speed, acceleration, state of charge, charging status, and power consumption data.
- Data preprocessing: It will involve preprocessing the collected data by performing data cleaning, filling missing values, selecting essential features, and normalizing the data.
- Machine learning model development: It will involve developing an Isolation Forest, One-Class SVM, and Deep Autoencoder model for identifying abnormal operating conditions.
- Anomaly detection: It will involve detecting the anomalies in the operational conditions of the electric vehicle.
- Fault severity classification: It will involve classifying the detected anomalies in terms of Low, Medium, High, and Critical using system-defined rules or a proper classification framework.
- Vehicle health assessment: It will involve providing the relevant vehicle health information along with the detected anomalies.
- Intelligent maintenance recommendation: It will provide maintenance recommendations based on the detected anomalies and their severity levels. Anomaly alerts: It will involve providing alerts about the detected anomalies.
- Historical analytics: It will store and manage the historical data of the detected anomalies and provide analytics on them.
- Mobile application: The proposed system will have a mobile application through which the end-users can view the vehicle health information, receive maintenance recommendations, get alerts about the anomalies, and view the analytic reports.
- Back-end and database: Python back-end with Fast API and RESTful API will be used for fulfilling the request of the mobile application. Firebase will be used for the user authentication and database management.

The proposed AI-Based EV Anomaly Detection System will be developed using Python and machine learning libraries including Scikit-learn, TensorFlow and Keras, Pandas, and NumPy for developing the prototype. These libraries will help in pre-processing the data, training and evaluating the machine learning model, and implementing the python back-end for the mobile application.

The technical aspects of the AI-Based EV Anomaly Detection System will be evaluated by testing whether the selected machine learning model can process the operational data, detect anomalies, and

provide results to the mobile application. Similarly, the technical results of the developed system will be presented using screenshots and test results of the implemented system.

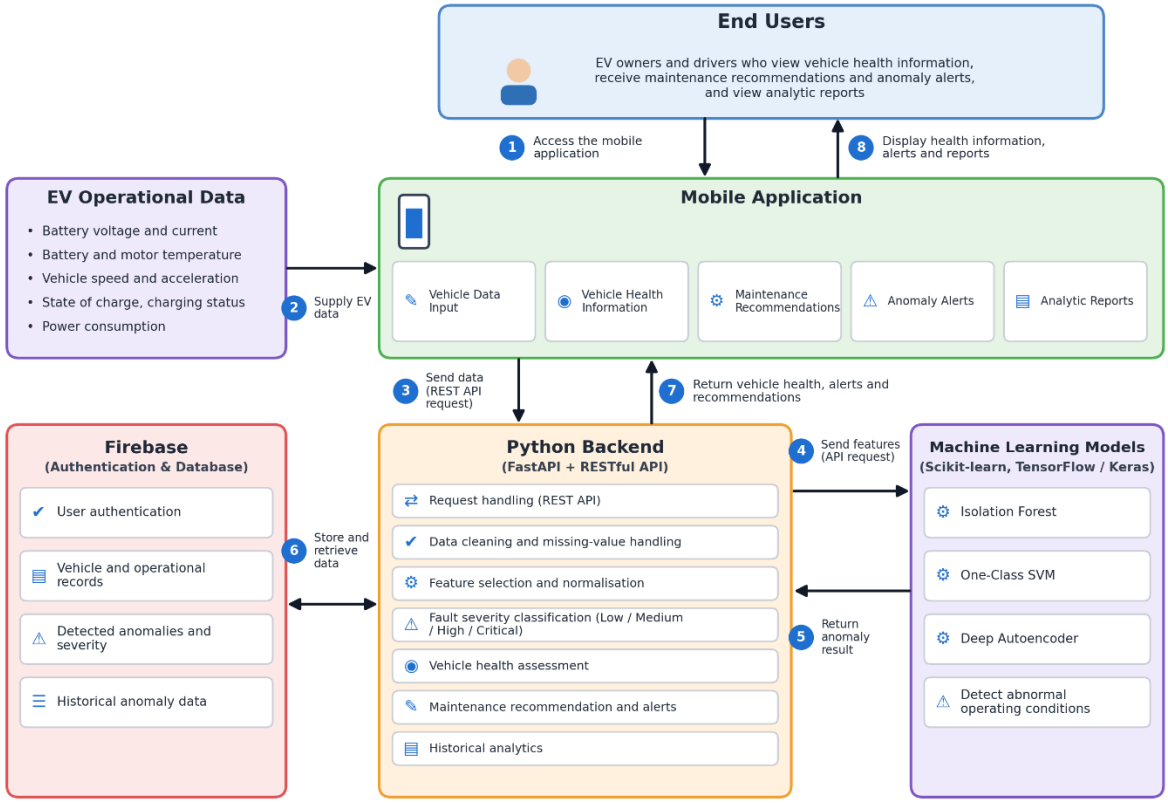


Figure 8 Technical Architecture of the AI-Based EV Anomaly Detection System

6.3 User or Stakeholder Validation

The proposed system is intended to cater to the needs of electric vehicle owners, fleet managers, service providers, and mechanics by offering information on abnormal vehicle operating conditions and maintenance recommendations.

The mobile application will offer vehicle health assessments, anomaly alerts, severity information of the faults, historical analytics, and maintenance recommendations, thus addressing the accessibility and interpretability of the information and supporting proactive maintenance decisions.

The following aspects will be subject to stakeholder validation:

- Usability: Whether the users will be able to navigate the app and access the primary functions without undue effort.

- Presentation: Whether the information on vehicle health assessments, anomaly alerts, and operating information will be effectively presented.
- Anomaly detection output: Whether the output indicating the abnormal conditions will be intuitive and informative.
- Severity: Whether the severity levels (Low, Medium, High, Critical) will be consistent with the indication of the detected faults.
- Maintenance recommendations: Whether the maintenance recommendations will be informative enough and address the detected irregularities.
- Historical analytics: Whether the users will be able to access historical data on anomalies and the evolution of operating conditions.
- Accessibility and security: Whether the users who have the appropriate credentials will be able to access the required information and whether the data about the vehicles is adequately protected.

The stakeholder validation can be performed by the target audience (e.g., EV owners or service persons) if the access to such people can be established. The validation method can include a questionnaire asking to rate the usability of the app, task-based tests asking to perform specific tasks, and an interview to identify broader improvement opportunities. The feedback will be compiled into a report highlighting the discovered issues, usability concerns, and areas for improvement.

The actual stakeholder validation results, including the number of participants and their feedback, will be available after the validation board conducts the necessary tests. Until the tests are done, these are the suggested areas for validation, not the actual results that the stakeholder analysis will bring.

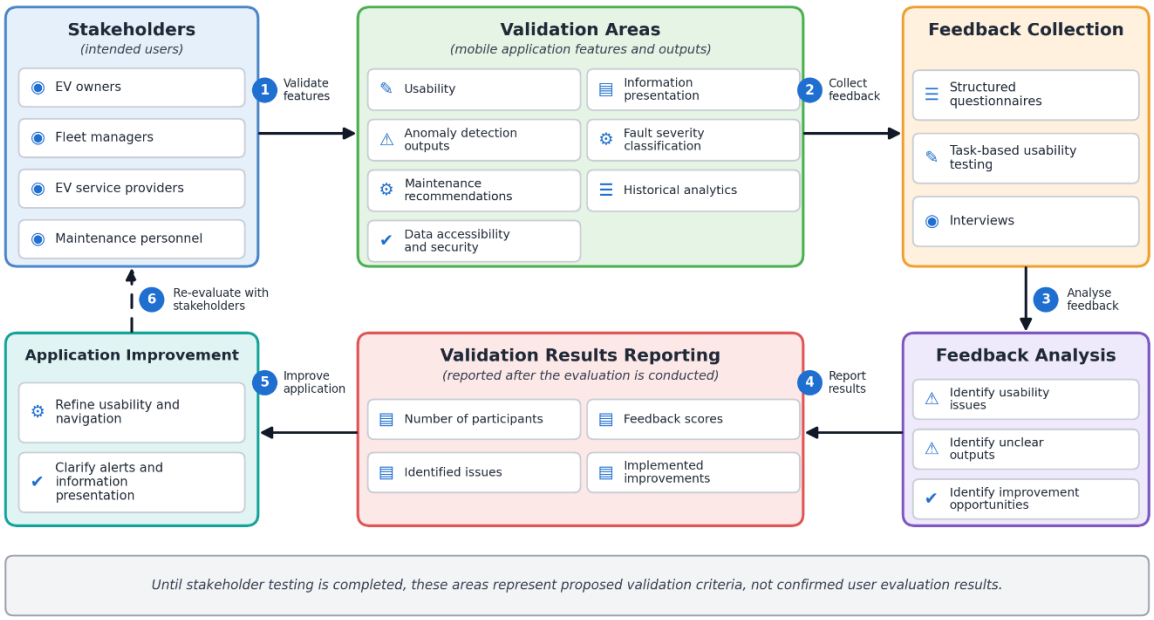


Figure 9 Stakeholder Validation Framework for the EV Anomaly Detection System

6.4 Integration Results

The proposed system architecture involves connecting an ML anomaly detection component to the Python backend, mobile application, and Firebase services. The aim of the integration is to ensure that the EV operational data is converted into information that can be accessed by a mobile interface.

The envisioned end-to-end workflow involves the following steps:

- Operational data preparation: The EV operational datasets involving battery, motor, speed, charging, and power parameters will be collected from public repositories and preprocessed for analysis
- Data preprocessing: The data will be preprocessed to remove missing and inconsistent values and unwanted features before it is fed into the ML models
- Anomaly detection: The ML model will detect anomalies in the processed dataset
- Result interpretation: The results will be interpreted to derive relevant vehicle health information and, if applicable, predict the fault severity based on the implemented rules
- Backend processing: The Python backend, powered by the Fast API web framework, will process incoming requests and return prediction results through RESTful APIs

- Database and authentication services: Firebase services will be used to provide the authentication and database services for the application, as described by the implemented data schema
- Mobile application presentation: The mobile application will present the results and alerts, as well as vehicle health and maintenance-related information to the users

The proposed system architecture will realize the intended connection between the data preprocessing and ML prediction modules, the Python backend, and the mobile interface through Firebase services. However, the extent to which the envisioned integration will take place will depend on the implementation and testing outcomes of the described components and their interrelations.

The integration will need to be validated by carrying out tests that will involve inputting valid and invalid entries and checking if API calls will return expected results. The following additional verification activities will need to be carried out to establish the level of successful integration: testing if the backend will respond to the prediction queries, if the response will include the predicted anomaly, if the backend errors will be handled correctly, if the database will be correctly read and written, if the authentication will be correctly handled, and if the results will be presented correctly in the mobile application. The response time of the prediction requests and the congruence of results between different components will also need to be evaluated.

The actual integration outcome should be documented with screenshots, API testing results, application results, and the relevant server log. It is important to note that components that have not been implemented or tested cannot be considered as integrated.

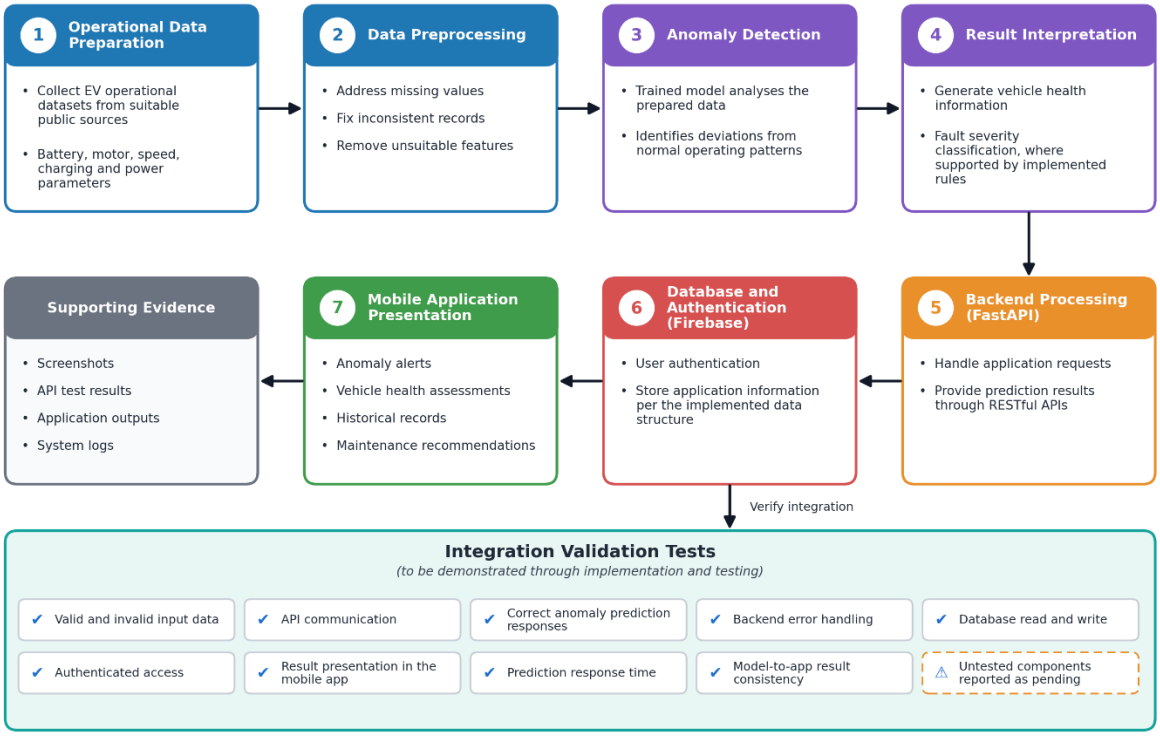


Figure 10 End-to-End Integration Workflow of the EV Anomaly Detection System

6.5 Summary of Findings

The research will propose an AI-Based EV Anomaly Detection System that will enhance the monitoring of electric vehicle operating conditions and enable predictive maintenance. Unlike traditional approaches to vehicle monitoring that use diagnostic trouble codes, the new system will utilize innovative machine learning algorithms that can detect irregularities in multiple operating parameters.

The research will compare the performance of three approaches to anomaly detection, including Isolation Forest, One-Class SVM, and Deep Autoencoder Neural Network. The performance of the three will be evaluated using precision, recall, F1-score, ROC-AUC where applicable, and prediction time. The comparison will help select the best-performing model for the proposed application.

The new system will be capable of detecting anomalies in operating conditions, providing vehicle health indicators, classifying the severity of detected faults, analyzing trends, and recommending predictive maintenance. The system will be built using a Python-based backend with Fast API framework to develop an API that will connect the machine learning module to the mobile application. Firebase will be used for user authentication and storing data.

The validation of the proposed system will involve testing the performance of the models, testing technical features, assessing system integration, analyzing data access, and validating the system's user

interface. After implementing the proposed system and completing tests, the final results, including stakeholder feedback, will inform the final conclusions.

In general, the proposed research will advance understanding of how AI and ML can be leveraged to monitor the operating conditions of electric vehicles and support predictive maintenance. Therefore, the final conclusions will be informed by the results of experiments and validation tests conducted to evaluate the performance of the proposed approaches.

7. DISCUSSION

7.1 Interpretation of Findings

The proposed Electric Vehicle Management System using Artificial Intelligence and Machine Learning is expected to enhance the monitoring of electric vehicles, battery management including charging, and user experience with the help of artificial intelligence and machine learning. The research study will implement AI/ML algorithms and a mobile application to offer insights into the performance of electric vehicles.

The proposed system will consist of four components, including AI-Based EV Operational Anomaly Detection, Adaptive Battery Health Prediction and Personalized Recommendation, Intelligent EV Charging Optimization and Range Prediction, and AI-Based EV Virtual Assistant. Each component will utilize the input data and address specific objectives, such as detecting EV operational anomalies, predicting battery health and providing recommendations, optimizing EV charging and predicting the range, and offering virtual assistance to the users, respectively.

The AI-Based EV Operational Anomaly Detection will focus on detecting various anomalies in the operational data of electric vehicles. The machine learning algorithms, including Isolation Forest, One-Class Support Vector Machine (One-Class SVM), and Deep Autoencoder Neural Networks, could be implemented to detect operational anomalies. The results could be offered through the mobile application to notify the users of the detected irregularities and recommend further actions.

The proposed system will be developed using React Native for the Mobile app, Python Fast API for the backend and AI/ML algorithms, and Firebase for the database and other related application services. The selected technologies will ensure that the developed mobile app will support the required communication with the backend, AI/ML algorithms, and database with the stored information on electric vehicles.

The proposed Electric Vehicle Management System using Artificial Intelligence and Machine Learning will conduct experiments and tests to evaluate the performance of the system. The experiments will report precision, recall, F1-score, anomaly detection rate, false-positive rate, battery health prediction error, and range prediction error for each component. However, since the experiments have not been conducted yet, the final results cannot be reported. In addition, the results could be generalized to the overall system to demonstrate the potential implementation and performance of AI/ML algorithms.

7.2 Comparison with Literature Baselines and Existing Solutions

Existing electric vehicle (EV) management approaches often provide features like battery diagnosis, charging information, driving range prediction, vehicle status reports, and maintenance recommendations. However, the nature and extent of such features depend significantly on the EV manufacturer, application, and monitoring tools. Conventional surveillance approaches provide little in the way of personalized recommendations or multi-intelligence features in a single application.

The proposed EV management system aims to address these challenges by integrating operational anomaly detection, battery health prediction, charging optimization, driving range prediction, and virtual assistant capabilities into a mobile-based platform. Rather than merely presenting information about the vehicle's status quo, the proposed system employs machine learning-based methodologies to detect irregularities and extract insights from the data.

For instance, the effectiveness of the proposed approach for operational anomaly detection can be compared to a suitable baseline method to evaluate whether such methods can help identify unusual behavior. Similarly, battery health and driving range prediction algorithms can be compared to relevant prediction-based approaches using relevant error metrics. Finally, charging optimization will be evaluated according to such factors as the estimated time-to-charge, efficiency, energy consumption rates, and the appropriateness of the suggested charging schedules.

The proposed system design involves developing an application using React Native and Python Fast API that will integrate the machine learning-based algorithms while Firebase will serve as the database. With the right implementation and testing, such a system would be able to offer an excellent end-user experience while also being easily maintained and updated.

However, a direct comparison to the existing solutions is not straightforward: for one, different approaches adopt different metrics, evaluation criteria, and experimentation conditions. As such, it is important to emphasize that the performance of the proposed system will not necessarily be better than the alternatives without an appropriate justification. Ultimately, the purpose of this comparison is to highlight the potential benefits and shortcomings of the integrated AI/ML-based approach while providing sufficient context for further experimentation.

7.3 Research Contribution

The primary research contribution involves the development of AI/ML-based features for electric vehicles. The proposed solution introduces the concept of an integrated system that combines multiple functions, including monitoring, analysis, charging support, and assistance.

The first contribution lies in the creation of an AI-based system for operational anomaly detection, which is designed to identify irregularities and notify the user. The second contribution is represented by an AI-based battery health predictor, which is intended to analyze the current state of the battery and offer recommendations.

The third contribution involves developing an intelligent system for optimizing battery charging and range prediction, which allows the user to plan the travel schedule and battery charging. The final research contribution includes the design of an AI-based virtual assistant for EVs, which enables hands-free control of the system and provides the user with relevant information.

Overall, the research involves the creation of an integrated solution that combines the use of React Native, python Fast API, Firebase, and machine learning algorithms. The proposed system can be used as a conceptual basis for future studies related to intelligent electric vehicles. Moreover, the current design can be expanded to include additional functions related to technical support and recommendation systems.

7.4 Product Contribution

The product contribution is an Electric Vehicle Management System conceptualized as a mobile application that will offer innovative approaches to managing electric vehicles. The application is an integration of a mobile frontend and backend, databases, and artificial intelligence and machine learning algorithms.

In this project, React Native was used to develop the application frontend, Python FastAPI was used to manage the application's communication with the machine learning models, and Firebase was used to manage databases and other application services.

The completed application consists of four major components, including

- AI-Based Electric Vehicle Operational Anomaly Detection which monitors the available data and notifies users of any irregularity.
- Adaptive Battery Health Prediction and Personalized Recommendation, another component, predicts the general health of the battery and offers personalized suggestions.
- The application also has an Intelligent EV Charging and Range Prediction component which manages and suggests the best way to charge the vehicle while also predicting the range of the electric vehicle.
- AI-Based EV Virtual Assistant which provides information and assistance to the application's users. By combining these features into a single application, developing this application will be highly innovative.

Moreover, it is possible to build upon the existing components to offer more data and make more predictions in the future.

7.5 Limitations and Threats to Validity

Several limitations can affect the proposed system's effectiveness. Firstly, machine learning algorithms' effectiveness relies on the amount and representativeness of the data used to train them. In this case, if the data samples do not represent different car models, battery conditions, driving behaviors, and environments, the results will not be accurate and generalizable.

Secondly, operational anomaly detection may result in false-positive or false-negative outcomes. These results can impact the accuracy of alerts and user's perception of their vehicle's condition. Thus, proper model assessment and threshold selection are crucial for this task.

Thirdly, the accuracy of battery health and driving range estimates may be impacted by batteries' aging, temperature, traffic, driving patterns, hilly roads, and energy consumption. If these variables are not considered during model training and evaluation, the system's accuracy will not be sufficient.

Moreover, the application's effectiveness relies on the stability of the connection between the user's device and the backend server, Firebase, or machine learning models. Poor connection, incorrect data input, or server errors can impact the performance of the mobile application. Data privacy and security should also be taken into account when storing or processing vehicle information.

Finally, the practical effectiveness of the proposed solution cannot be determined during implementation. Correctness of the chosen approach, model evaluation, and system integration should be validated during the testing and assessment phase. The described system is designed to assist users in understanding their vehicle's condition; thus, it should not affect official car inspection and manufacturer's recommendations.

7.6 Implications

The proposed system could potentially benefit electric vehicle owners, vehicle maintenance, charging control, and advanced transportation technologies. The application can aid users in comprehending vehicle status and support them in making informed battery maintenance and charging decisions via the implemented monitoring and prediction mechanisms.

The detection of abnormal vehicle operations can contribute to identifying irregularities during the driving process and determining whether the issue requires further inspection. The suggested battery prediction mechanism and personalized suggestion system could empower users with a better perception of their vehicle's battery state. Moreover, the proposed charging control and range prediction functions could assist in estimating the driving range and ensuring convenient charging station visits.

In terms of technology, the proposed system presents a viable option for developing an advanced mobile application that supports diverse operations for managing electric vehicles. The utilization of the React Native framework, Python Fast API, Firebase, and various artificial intelligence and machine learning solutions indicate the developer's capacity to address the challenges associated with building such a system. At the same time, the modular structure of the designed solution enables the enhancement of the system's capabilities and the addition of new functions.

From a research perspective, the proposed system could be employed to compare the performance of different machine learning algorithms and enhance the accuracy of the predictions provided. The validity of personalization as an approach to increasing user engagement and satisfaction could be verified. However, achieving positive outcomes depends on the quality of the data utilized, the implementation details, and the correctness of the employed mathematical approaches.

8COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

The Electric Vehicle Management System using AI and ML will offer intelligent vehicle insights, battery health and charging analytics, driving range prediction, and a virtual assistant. The proposed system will comprise four major modules to enable EV owners to operate their vehicles efficiently.

The primary target users of the proposed system are EV owners, drivers, and charging station providers. In addition, businesses that provide fleet management services for electric vehicles will benefit from the proposed system. The system will leverage the capabilities of AI/ML and the application's interface to offer insights into electric vehicle's performance. Nonetheless, the proposed venture's feasibility needs to be evaluated to ensure that it addresses the intended users' needs.

8.2 Market and Competitive Position

The rise of electric vehicles presents an excellent opportunity to develop innovative solutions for EV owners, drivers, and providers. The current applications offer insights into the vehicle's battery, charging processes, and general performance depending on the manufacturer. However, the proposed system intends to offer competitive advantages over the current solutions by integrating the operational insights, battery health and charging analytics, driving range prediction, and a virtual assistant in one platform, thus improving convenience and accessibility. React Native framework will be used to develop the application while Fast API, a Python web framework, will handle the backend. In addition, Firebase will be used as a database and application hosting platform.

While the proposed system might offer competitive advantages, its position in the market needs to be evaluated, including user feedback and the availability of alternative solutions for managing EVs.

8.3 Business or Sustainability Model

The Electric Vehicle Management System could generate income from individual EV owners and businesses through subscriptions to the services provided by the application. In addition, the system could offer fleet management solutions to business owners, including commission on sales, premium subscription plans, and maintenance services for the application. The system could also offer enhanced AI-powered features to increase revenues from its target users.

Sustainability will be achieved through continuous improvements to the application, including updating the AI/ML algorithms and database to keep up with the evolving technology and emerging trends. In addition, the system will rely on user feedback to enhance existing features and add new ones. However, the Electric Vehicle Management System will need to consider other aspects of a strong business model,

including cost and sustainability of the infrastructure, and the accessibility of the proposed solutions. As such, this venture's commercial potential still needs to be investigated.

8.4 Scalability, Intellectual Property and Compliance

The proposed system can be scaled up to serve a large population of EV owners and drivers. It can also be modified to support different EV models and manufacturers. Additionally, the system can be adjusted to meet the needs of business owners offering fleet management services in the future. The system also boasts modular components that allow for the addition of new features and functions as needed. In addition, the system can consider using existing AI/ML algorithms and datasets to enhance its performance.

As the creator of the proposed system, one has to ensure that the work done is free from copyright violations and intellectual property issues. The source code, ML algorithms, datasets, and other third-party tools used to develop the application should respect their licensing terms and conditions. Moreover, the system should have updated security features to protect the data and information of the users from illegal access and manipulation. The system's predictions should not be regarded as diagnostic tools but rather as suggestions that the users can choose to adopt or not. It is also essential to conduct tests and quality assurance of the developed application before it is deployed. In addition, the system must comply with the relevant data protection and consumer protection regulations before launch.

8.5 Contribution to the Sustainable Development Goals

The proposed Electric Vehicle Management System using AI and ML has the potential to contribute to several of the United Nations Sustainable Development Goals (SDGs). The system can contribute to the following:

- SDG 7 – Affordable and Clean Energy – Through its battery health and charging analytics feature, the system will help promote the production and consumption of clean energy.
- SDG 9 – Industry, Innovation, and Infrastructure – The application will foster innovation and infrastructural development by creating an ecosystem for implementing state-of-the-art technology in EV management.
- SDG 11 – Sustainable Cities and Communities – The system will play a role in promoting sustainable transportation, which will lead to the creation of sustainable cities and communities.

- SDG 12 – Responsible Consumption and Production – By encouraging users to adopt energy-efficient practices, such as purchasing electric vehicles, the system will promote responsible consumption and production of energy.

The potential contributions of the proposed system to the UN SDGs are discussed, and the extent to which they will be achieved depends on the system's adoption rate and overall impact.

9. CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

The proposed Electric Vehicle Management System utilizing AI/ML is expected to enhance the monitoring, battery maintenance, charging optimization, and user assistance processes in electric vehicles. The proposed system comprises four modules, namely, AI-Based EV Operational Anomaly Detection, Adaptive Battery Health Prediction and Personalized Recommendation, Intelligent EV Charging Optimization and Range Prediction, and AI-Based EV Virtual Assistant.

The researchers intend to utilize React Native for mobile application development, Python FastAPI framework for the back-end, and Firebase to empower the application with various services. The AI/ML techniques are expected to enable the system carry out anomaly detection, battery health prediction, driving range prediction, and recommend personalized EV options.

Notably, the proposed system aims to provide an integrated mobile application that will combine multiple functionalities related to EV management, operation, and user assistance. While the above qualities are expected to enhance the performance of the system, it remains crucial to conduct experiments and tests to verify the actual performance of the proposed system.

9.2 Achievement of Objectives

The proposed Electric Vehicle Management System will achieve its objectives through the development of four modules, mentioned above. The first objective of the proposed system is to detect operational anomalies in EVs, which will be achieved through the utilization of AI/ML-based anomaly detection procedures. The second objective is to predict battery health and provide personalized recommendations, which will be achieved through the utilization of the Adaptive Battery Health Prediction and Personalized Recommendation module.

The third objective is to enable intelligent EV charging optimization and driving range prediction, which will be achieved through the application of the Intelligent EV Charging Optimization and Range Prediction module. Finally, the fourth objective will be achieved through the development of an AI-Based EV Virtual Assistant that will provide EV-related information and explicate the system's recommendations.

The achievement of the objectives will be realized through the combination of React Native, Python Fast API, and Firebase to form an integrated system for mobile application-based EV management. The accomplishment of the aforementioned objectives should be measurable and verifiable through

implementational experiments and objective analyses such as calculating the relevant performance metrics of machine learning algorithms.

9.3 Individual Research and Product Contributions

The research component of the proposed Electric Vehicle Management System aims to integrate AI/ML to develop functionalities that will enhance EV monitoring, battery maintenance, charging optimization, and user assistance. The research will elaborate on the potential of utilizing the characteristics of various EVs to create predictive and recommendatory models that can be used to predict operational anomalies, driving range, and battery health, among other applications.

The product component entails the development of a mobile application that will combine functionalities related to EV management, operation, and user assistance. Specifically, the product component will entail the development of an integrated application that will combine the four aforementioned modules to form an integrated system capable of providing information, making predictions, and offering recommendations.

The proposed research and product components make a significant contribution toward the development of intelligent EV management systems. Notably, the proposed application will set the foundation for the development of mobile application-based intelligent systems in general and specifically in the context of EV management. The contribution will be achieved through the combination of the four functionalities into a single mobile application.

The practical implementation and verifiability of the proposed application will depend on various factors, including the performance of the models and the system's overall performance. Additionally, the application will require extensive testing to ensure that it is intuitive and that it meets the needs of the intended users.

9.4 Future Directions

Numerous future directions can be considered in the context of the proposed Electric Vehicle Management System. First, the proposed system can be evaluated using real-world data to ascertain its applicability and effectiveness. The evaluation process will entail collecting and utilizing data from numerous EV models to assess the performance of the system's modules under diverse conditions.

Second, the feasibility of the proposed application can be evaluated by comparing the performance of diverse machine learning algorithms. The performance of the anomaly detection component can be compared using performance metrics such as precision, recall, F1-score, and false-positive rates.

Similarly, the prediction performance of the battery health and range prediction components can be compared using relevant performance metrics.

The driving range component can be enhanced by taking into account the time and location of the driver, the distance of the journey, and the availability and accessibility of charging stations. The personalization aspect can be considered further to enhance the personalization features based on user preferences and the historical data of the user's vehicle.

The AI-Based EV Virtual Assistant can be considered further to offer more sophisticated and intuitive interactions and provide explication of the recommendations and predictions. Future considerations can focus on achieving application security and usability, among other qualities. Generally, future research can focus on the pilot testing of the application and the identification of the aspects that require further improvement.

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GLOSSARY

Anomaly Detection refers to the discovery of unusual patterns or observations that deviate from typical operations of electric vehicle data.

Autoencoder is a neural network that learns a reconstruction function of the input data, allowing to find possible anomalies.

Battery State of Charge (SoC) estimates how much electricity a battery can provide as a percentage of its total capacity.

Feature Extraction denotes transformation of raw electric vehicle measurements into a set of descriptive features that can be recognized by a machine learning algorithm.

Isolation Forest is an isolation-based algorithm used to detect anomalies.

Machine Learning Model is an algorithm that is trained on a particular set of data in order to find some patterns in data or classify observations in general.

One-Class Support Vector Machine (One-Class SVM) is an unsupervised outlier detection algorithm that learns the support vectors from normal observations.

Predictive Maintenance aims to find ways to predict possible equipment failure and perform maintenance activities to avoid serious malfunctions.

Statistical Features are the measurements that are used to estimate a certain characteristic of a statistical population, such as mean, minimum, maximum, standard deviation, etc.

Vehicle Health Assessment is a procedure that determines a vehicle's health condition based on available measurements and found anomalies.

Anomaly Severity Classification performs classification of found anomalies according to some specific criteria.

Real-Time Monitoring implies the delivery of monitoring data to users with the delay that depends on the message size and transmission speed.

Application Programming Interface (API) represents an application programming interface that allows to access specific features of a web service, for instance, to perform a request in order to get information about a web app from the service.

Backend is the server part of the app that processes requests and usually contains the database and algorithms, such as machine learning models, for instance, for data processing.

Dataset denotes a set of data samples for the machine learning pipeline.

APPENDICES

Appendix A: Supplementary Research and Evaluation Information

This appendix aims to provide further information related to the development and evaluation of the proposed AI-Based Electric Vehicle Anomaly Detection System. In particular, this section will discuss the operational data that will be utilized by the system, machine learning evaluation results, and additional software test cases.

A.1 Electric Vehicle Data Collection and Preprocessing

The proposed system will utilize electric vehicle operational data either from the selected dataset or data source. The operational parameters that will be considered depend on the selected data source and may involve battery voltage, battery current, battery temperature, motor temperature, vehicle speed, state of charge, acceleration, charging status, power consumption, etc.

Such data will be checked and preprocessed before being fed into the machine learning models. The preprocessing stage may involve removing missing values, checking invalid measurements, converting the data to numerical values, and extracting additional features, if required.

The following features can be extracted from the available data:

- Average battery voltage and current
- minimum and maximum battery temperature
- average and maximum motor temperature
- average vehicle speed and acceleration
- battery state of charge: power consumption statistics
- standard deviation of the selected measurements

Only the features supported by the selected data source and implemented preprocessing procedure should be included in the final report.

A.2 Machine Learning Evaluation Results

Machine learning model performance should be discussed using the results obtained from the implemented experiments. The following table provides a template for presenting the evaluation results.

Table 10 Evaluation Matrix

Evaluation Metric	**Result**
Accuracy	93.75 %

Balanced accuracy	92.50 %
Macro precision	93.34 %
Macro recall	92.50 %
Macro F1-score	92.50 %
Weighted F1-score	93.72 %

The evaluation should compare the performance of the considered machine learning models, including Isolation Forest, One-Class SVM, and a deep autoencoder, where applicable. The final results should be obtained from the actual model evaluation procedure rather than estimated or invented.

A.3 Additional Software Test Cases

Table 11 Test case

Test ID	Test Description	Expected Result	Actual Result
AT01	Load or import EV operational data	Data are loaded successfully from the configured source	To be completed
AT02	Validate input measurements	Valid measurements are accepted and invalid data are handled appropriately	To be completed
AT03	Preprocess operational data	Data are transformed into the required input format	To be completed
AT04	Extract machine learning features	Required features are calculated correctly	To be completed
AT05	Generate an anomaly prediction	The configured model returns a prediction	To be completed
AT06	Classify an abnormal operating condition	The system assigns a severity level if this functionality is implemented	To be completed
AT07	Display vehicle health results	Prediction results are displayed correctly in the mobile application	To be completed

AT08	Store prediction history	Prediction information is saved correctly if historical storage is implemented	To be completed
AT09	Generate a maintenance recommendation	A recommendation is produced according to the implemented rules or model	To be completed
AT10	Handle backend or communication errors	The system handles failures appropriately and provides suitable feedback	To be completed

Appendix B: Academic Integrity and AI Use Disclosure

B.1 Purpose of AI Use

AI-based tools were used as auxiliary support while preparing this report and may have provided assistance with report organization, academic wording, and technical explanations. The research design, software implementation, experiments, results, and academic claims should reflect the actual work performed and be verified by the researcher.

B.2 Tools Used

Table 12 Tools

Tool	**Purpose**
ChatGPT	Assisted with report organization, academic wording, and technical explanations, and preliminary descriptions of the proposed system.
Google Colab	Model development, training, and evaluation, if used during the documented experiments.
Visual Studio Code	Software development and code editing.
Postman	Backend API testing

Git/GitHub	Source code, version control and project collaboration
Scikit-learn	Implementation and evaluation of machine learning algorithms
TensorFlow/Keras	Development and evaluation of neural network models

Only the tools actually used in the project should be retained in the final disclosure.